

OVP Guide to Using Processor Models

Model specific information for ARM_Cortex-A76MPx4

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Chapter 1

Overview

This document provides the details of an OVP Fast Processor Model variant.

OVP Fast Processor Models are written in C and provide a C API for use in C based platforms. The models also provide a native interface for use in SystemC TLM2 platforms.

The models are written using the OVP VMI API that provides a Virtual Machine Interface that defines the behavior of the processor. The VMI API makes a clear line between model and simulator allowing very good optimization and world class high speed performance. Most models are provided as a binary shared object and also as source. This allows the download and use of the model binary or the use of the source to explore and modify the model.

The models are run through an extensive QA and regression testing process and most model families are validated using technology provided by the processor IP owners. There is a companion document (OVP Guide to Using Processor Models) which explains the general concepts of OVP Fast Processor Models and their use. It is downloadable from the OVPworld website documentation pages.

1.1 Description

ARM Processor Model

1.2 Licensing

Usage of binary model under license governing simulator usage.

Note that for models of ARM CPUs the license includes the following terms:

Licensee is granted a non-exclusive, worldwide, non-transferable, revocable licence to:

If no source is being provided to the Licensee: use and copy only (no modifications rights are granted) the model for the sole purpose of designing, developing, analyzing, debugging, testing, verifying, validating and optimizing software which: (a) (i) is for ARM based systems; and (ii) does not incorporate the ARM Models or any part thereof; and (b) such ARM Models may not be used

to emulate an ARM based system to run application software in a production or live environment.

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In the case of any Licensee who is either or both an academic or educational institution the purposes shall be limited to internal use.

Except to the extent that such activity is permitted by applicable law, Licensee shall not reverse engineer, decompile, or disassemble this model. If this model was provided to Licensee in Europe, Licensee shall not reverse engineer, decompile or disassemble the Model for the purposes of error correction.

The License agreement does not entitle Licensee to manufacture in silicon any product based on this model.

The License agreement does not entitle Licensee to use this model for evaluating the validity of any ARM patent.

Source of model available under separate Imperas Software License Agreement.

1.3 Limitations

Instruction pipelines are not modeled in any way. All instructions are assumed to complete immediately. This means that instruction barrier instructions (e.g. ISB, CP15ISB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. The model does not implement speculative fetch behavior. The branch cache is not modeled.

Caches and write buffers are not modeled in any way. All loads, fetches and stores complete immediately and in order, and are fully synchronous (as if the memory was of Strongly Ordered or Device-nGnRnE type). Data barrier instructions (e.g. DSB, CP15DSB) are treated as NOPs, with the exception of any undefined instruction behavior, which is modeled. Cache manipulation instructions are implemented as NOPs, with the exception of any undefined instruction behavior, which is modeled.

Real-world timing effects are not modeled: all instructions are assumed to complete in a single cycle.

Performance Monitors are implemented as a register interface only except for the cycle counter, which is implemented assuming one instruction per cycle.

TLBs are architecturally-accurate but not device accurate. This means that all TLB maintenance and address translation operations are fully implemented but the cache is larger than in the real device.

DynamIQ Shared Unit Control registers are implemented as a register interface only - caches and busses are not modeled.

DynamIQ Shared Unit Performance Monitor registers are implemented as a register interface only

- caches and busses are not modeled.

Activity Monitor registers are implemented as a register interface only.

Debug registers are implemented but non-functional (which is sufficient to allow operating systems such as Linux to boot). Debug state is not implemented.

The GICv3 block is implemented without any ITS. Implementation-defined GICR registers for control of LPIs (GICR_SETLPIR, GICR_CLRLPIR, GICR_INVLPIR, GICR_INVALLR and GICR_SYNCR) are all implemented.

1.4 Verification

Models have been extensively tested by Imperas. ARM Cortex-A models have been successfully used by customers to simulate SMP Linux, Ubuntu Desktop, VxWorks and ThreadX on Xilinx Zynq virtual platforms.

1.5 Features

The precise set of implemented features in the model is defined by ID registers. Use overrides to modify these if required (for example `override_PFR0` or `override_AA64PFR0_EL1`).

1.5.1 Core Features

AArch64 is implemented at EL3, EL2, EL1 and EL0.

AArch32 is implemented at EL1 and EL0.

The following ARMv8.0 core features are implemented: FEAT_CSV2, FEAT_CSV3, FEAT_SSBS.

The following ARMv8.0 cryptographic features are implemented: FEAT_AES, FEAT_PMULL, FEAT_SHA1, FEAT_SHA256.

The following ARMv8.1 core features are implemented: FEAT_HAFDBS (access flag and dirty state), FEAT_HPDS, FEAT_LOR, FEAT_LSE, FEAT_PAN, FEAT_RDM, FEAT_VHE, FEAT_VMID16.

The following ARMv8.2 core features are implemented: FEAT_AA32HPD, FEAT_DotProd, FEAT_DPB, FEAT_FP16, FEAT_HPDS2, FEAT_IESB, FEAT_PAN2, FEAT_RAS (with no error records), FEAT_TTCNP, FEAT_UAO, FEAT_XNX.

The following ARMv8.3 core features are implemented: FEAT_LRCPC.

1.5.2 Memory System

Security extensions are implemented (also known as TrustZone). To make non-secure accesses visible externally, override `ID_AA64MMFR0_EL1.PARange` to specify the required physical bus size (32, 36, 40, 42, 44, 48 or 52 bits) and connect the processor to a bus one bit wider (33, 37,

41, 43, 45, 49 or 53 bits, respectively). The extra most-significant bit is the NS bit, indicating a non-secure access. If non-secure accesses are not required to be made visible externally, connect the processor to a bus of exactly the size implied by ID_AA64MMFR0_EL1.PARange.

VMSA EL1, EL2 and EL3 stage 1 address translation is implemented. VMSA stage 2 address translation is implemented.

LPA (large physical address extension) is implemented as standard in ARMv8.

TLB behavior is controlled by parameter ASIDCacheSize. If this parameter is 0, then an unlimited number of TLB entries will be maintained concurrently. If this parameter is non-zero, then only TLB entries for up to ASIDCacheSize different ASIDs will be maintained concurrently initially; as new ASIDs are used, TLB entries for less-recently used ASIDs are deleted, which improves model performance in some cases (especially when 16-bit ASIDs are in use). If the model detects that the TLB entry cache is too small (entry ejections are very frequent), it will increase the cache size automatically. In this variant, ASIDCacheSize is 8

1.5.3 Advanced SIMD and Floating-Point Features

SIMD and VFP instructions are implemented.

The model implements trapped exceptions if FPTrap is set to 1 in MVFR0 (for AArch32) or MVFR0_EL1 (for AArch64). When floating point exception traps are taken, cumulative exception flags are not updated (in other words, cumulative flag state is always the same as prior to instruction execution, even for SIMD instructions). When multiple enabled exceptions are raised by a single floating point operation, the exception reported is the one in least-significant bit position in FPSCR (for AArch32) or FPCR (for AArch64). When multiple enabled exceptions are raised by different SIMD element computations, the exception reported is selected from the lowest-index-number SIMD operation. Contact Imperas if requirements for exception reporting differ from these.

Trapped exceptions not are implemented in this variant (FPTrap=0)

1.5.4 Generic Timer

Generic Timer is present. Use parameter “override_timerScaleFactor” to specify the counter rate as a fraction of the processor MIPS rate (e.g. 10 implies Generic Timer counters increment once every 10 processor instructions).

1.5.5 Generic Interrupt Controller

GIC block is implemented (GICv3, including security extensions). Accesses to GIC registers can be viewed externally by connecting to the 32-bit GICRegisters and GICDRegisters bus ports. Secure register accesses are at offset 0x0 on these busses; for example, a secure access to GICD register GICD_CTLR can be observed by monitoring address 0x00000000 of bus GICDRegisters. Non-secure accesses are at offset 0x80000000 on these busses; for example, a non-secure access to GICD register GICD_CTLR can be observed by monitoring address 0x80000000 of bus GICDRegisters

GIC Distributor registers are located at address 0x2f000000. Use parameter “override_GICv3_DistributorBase” to change this if required.

The internal GIC block can be disabled by raising signal GICCDISABLE, in which case the GIC needs to be modeled using a platform component instead. Input signals `vfiq_CPU<N>` and `virq_CPU<N>` can be used by this component to raise virtual FIQ and IRQ interrupts on cores in the cluster if required.

1.6 Debug Mask

It is possible to enable model debug features in various categories. This can be done statically using the “`override_debugMask`” parameter, or dynamically using the “`debugflags`” command. Enabled debug features are specified using a bitmask value, as follows:

Value 0x004: enable debugging of MMU/MPU mappings.

Value 0x020: enable debugging of reads and writes of GIC block registers.

Value 0x040: enable debugging of exception routing via the GIC model component.

Value 0x080: enable debugging of all system register accesses.

Value 0x100: enable debugging of all traps of system register accesses.

Value 0x200: enable verbose debugging of other miscellaneous behavior (for example, the reason why a particular instruction is undefined).

Value 0x400: enable debugging of Performance Monitor timers

Value 0x800: enable dynamic validation of TLB entries against in-memory page table contents (finds some classes of error where page table entries are updated without a subsequent flush of affected TLB entries).

All other bits in the debug bitmask are reserved and must not be set to non-zero values.

1.7 AArch32 Unpredictable Behavior

Many AArch32 instruction behaviors are described in the ARM ARM as **CONSTRAINED UNPREDICTABLE**. This section describes how such situations are handled by this model.

1.7.1 Equal Target Registers

Some instructions allow the specification of two target registers (for example, double-width `SMULL`, or some `VMOV` variants), and such instructions are **CONSTRAINED UNPREDICTABLE** if the same target register is specified in both positions. In this model, such instructions are treated as **UNDEFINED**.

1.7.2 Floating Point Load/Store Multiple Lists

Instructions that load or store a list of floating point registers (e.g. `VSTM`, `VLDM`, `VPUSH`, `VPOP`) are **CONSTRAINED UNPREDICTABLE** if either the uppermost register in the specified

range is greater than 32 or (for 64-bit registers) if more than 16 registers are specified. In this model, such instructions are treated as UNDEFINED.

1.7.3 Floating Point VLD[2-4]/VST[2-4] Range Overflow

Instructions that load or store a fixed number of floating point registers (e.g. VST2, VLD2) are CONSTRAINED UNPREDICTABLE if the upper register bound exceeds the number of implemented floating point registers. In this model, these instructions load and store using modulo 32 indexing (consistent with AArch64 instructions with similar behavior).

1.7.4 If-Then (IT) Block Constraints

Where the behavior of an instruction in an if-then (IT) block is described as CONSTRAINED UNPREDICTABLE, this model treats that instruction as UNDEFINED.

1.7.5 Use of R13

In architecture variants before ARMv8, use of R13 was described as CONSTRAINED UNPREDICTABLE in many circumstances. From ARMv8, most of these situations are no longer considered unpredictable. This model allows R13 to be used like any other GPR, consistent with the ARMv8 specification.

1.7.6 Use of R15

Use of R15 is described as CONSTRAINED UNPREDICTABLE in many circumstances. This model allows such use to be configured using the parameter “unpredictableR15” as follows:

Value “undefined”: any reference to R15 in such a situation is treated as UNDEFINED;

Value “nop”: any reference to R15 in such a situation causes the instruction to be treated as a NOP;

Value “raz_wi”: any reference to R15 in such a situation causes the instruction to be treated as a RAZ/WI (that is, R15 is read as zero and write-ignored);

Value “execute”: any reference to R15 in such a situation is executed using the current value of R15 on read, and writes to R15 are allowed (but are not interworking).

Value “assert”: any reference to R15 in such a situation causes the simulation to halt with an assertion message (allowing any such unpredictable uses to be easily identified).

In this variant, the default value of “unpredictableR15” is “undefined”.

1.7.7 Unpredictable Instructions in Some Modes

Some instructions are described as CONSTRAINED UNPREDICTABLE in some modes only (for example, MSR accessing SPSR is CONSTRAINED UNPREDICTABLE in User and System

modes). This model allows such use to be configured using the parameter “unpredictableModal”, which can have values “undefined” or “nop”. See the previous section for more information about the meaning of these values.

In this variant, the default value of “unpredictableModal” is “undefined”.

1.8 Integration Support

This model implements a number of non-architectural pseudo-registers and other features to facilitate integration.

1.8.1 Memory Transaction Query

Three registers are intended for use within memory callback functions to provide additional information about the current memory access. Register `transactPL` indicates the processor execution level of the current access (0-3). Note that for load/store translate instructions (e.g. `LDRT`, `STRT`) the reported execution level will be 0, indicating an EL0 access. Register `transactAT` indicates the type of memory access: 0 for a normal read or write; and 1 for a physical access resulting from a page table walk. Register `atomicType` indicates whether the current instruction has any special atomic constraints (0 indicates no constraint, 1 indicates atomic, 2 indicates exclusive access).

1.8.2 Page Table Walk Query

A banked set of registers provides information about the most recently completed page table walk. There are up to six banks of registers: bank 0 is for stage 1 walks, bank 1 is for stage 2 walks, and banks 2-5 are for stage 2 walks initiated by stage 1 level 0-3 entry lookups, respectively. Banks 1-5 are present only for processors with virtualization extensions. The currently active bank can be set using register `PTWBankSelect`. Register `PTWBankValid` is a bitmask indicating which banks contain valid data: for example, the value 0xb indicates that banks 0, 1 and 3 contain valid data.

Within each bank, there are registers that record addresses and values read during that page table walk. Register `PTWBase` records the table base address, register `PTWInput` contains the input address that starts a walk, register `PTWOutput` contains the result address and register `PTWPgSize` contains the page size (`PTWOutput` and `PTWPgSize` are valid only if the page table walk completes). Registers `PTWAddressL0`-`PTWAddressL3` record the addresses of level 0 to level 3 entries read, respectively. Register `PTWAddressValid` is a bitmask indicating which address registers contain valid data: bits 0-3 indicate `PTWAddressL0`-`PTWAddressL3`, respectively, bit 4 indicates `PTWBase`, bit 5 indicates `PTWInput`, bit 6 indicates both `PTWOutput` and `PTWPgSize`. For example, the value 0x73 indicates that `PTWBase`, `PTWInput`, `PTWOutput`, `PTWPgSize` and `PTWAddressL0`-`L1` are valid but `PTWAddressL2`-`L3` are not. Register `PTWAddressNS` is a bitmask indicating whether an address is in non-secure memory: bits 0-3 indicate `PTWAddressL0`-`PTWAddressL3`, respectively, bit 4 indicates `PTWBase`, bit 6 indicates `PTWOutput` (`PTWInput` is a VA and thus has no secure/non-secure info). Registers `PTWValueL0`-`PTWValueL3` contain page table entry values read at level 0 to level 3. Register `PTWValueValid` is a bitmask indicating which value registers contain valid data: bits 0-3 indicate `PTWValueL0`-`PTWValueL3`, respectively.

1.8.3 Artifact Page Table Walks

Registers are also available to enable a simulation environment to initiate an artifact page table walk (for example, to determine the ultimate PA corresponding to a given VA). Register PTWI_EL1S initiates a secure EL1 table walk for a fetch. Register PTWD_EL1S initiates a secure EL1 table walk for a load or store (note that current ARM processors have unified TLBs, so these registers are synonymous). Registers PTW[ID]_EL1NS initiate walks for non-secure EL1 accesses. Registers PTW[ID]_EL2 initiate EL2 walks. Registers PTW[ID]_S2 initiate stage 2 walks. Registers PTW[ID]_EL3 initiate AArch64 EL3 walks. Finally, registers PTW[ID]_current initiate current-mode walks (useful in a memory callback context). Each walk fills the query registers described above.

1.8.4 MMU and Page Table Walk Events

Two events are available that allow a simulation environment to be notified on MMU and page table walk actions. Event mmuEnable triggers when any MMU is enabled or disabled. Event pageTableWalk triggers on completion of any page table walk (including artifact walks).

1.8.5 Artifact Address Translations

A simulation environment can trigger an artifact address translation operation by writing to the architectural address translation registers (e.g. ATS1CPR). The results of such translations are written to an integration support register artifactPAR, instead of the architectural PAR register. This means that such artifact writes will not perturb architectural state.

1.8.6 TLB Invalidation

A simulation environment can cause TLB state for one or more address translation regimes in the processor to be flushed by writing to the artifact register ResetTLBs. The argument is a bitmask value, in which non-zero bits select the TLBs to be flushed, as follows:

Bit 0: EL0/EL1 stage 1 secure TLB

Bit 1: EL0/EL1 stage 1 non-secure TLB

Bit 2: EL2 stage 1 non-secure TLB

Bit 3: EL0/EL1 stage 2 non-secure TLB

Bit 4: EL3 stage 1 TLB

1.8.7 Halt Reason Introspection

An artifact register HaltReason can be read to determine the reason or reasons that a processor is halted. This register is a bitfield, with the following encoding: bit 0 indicates the processor has executed a wait-for-event (WFE) instruction; bit 1 indicates the processor has executed a wait-for-interrupt (WFI) instruction; and bit 2 indicates the processor is held in reset.

1.8.8 Automatic WFI Wake Up

A simulation environment can cause a core to wake up from the WFI state by writing to the artifact register WakeWFINow.

1.8.9 System Register Access Monitor

If parameter “enableSystemMonitorBus” is True, an artifact 32-bit bus “SystemMonitor” is enabled for each PE. Every system register read or write by that PE is then visible as a read or write on this artifact bus, and can therefore be monitored using callbacks installed in the client environment (use opBusReadMonitorAdd/opBusWriteMonitorAdd or icmAddBusReadCallback-/icmAddBusWriteCallback, depending on the client API). The format of the address on the bus is as follows:

bits 31:26 - zero

bit 25 - 1 if AArch64 access, 0 if AArch32 access

bit 24 - 1 if non-secure access, 0 if secure access

bits 23:20 - CRm value

bits 19:16 - CRn value

bits 15:12 - op2 value

bits 11:8 - op1 value

bits 7:4 - op0 value (AArch64) or coprocessor number (AArch32)

bits 3:0 - zero

As an example, to view non-secure writes to writes to CNTFRQ_EL0 in AArch64 state, install a write monitor on address range 0x020e0330:0x020e0333.

1.8.10 System Register Implementation

If parameter “enableSystemBus” is True, an artifact 32-bit bus “System” is enabled for each PE. Slave callbacks installed on this bus can be used to implement modified system register behavior (use opBusSlaveNew or icmMapExternalMemory, depending on the client API). The format of the address on the bus is the same as for the system monitor bus, described above.

Chapter 2

Configuration

2.1 Location

This model's VLVN is `arm.ovpworld.org/processor/arm/1.0`.

The model source is usually at:

`$IMPERAS_HOME/ImperasLib/source/arm.ovpworld.org/processor/arm/1.0`

The model binary is usually at:

`$IMPERAS_HOME/lib/$IMPERAS_ARCH/ImperasLib/arm.ovpworld.org/processor/arm/1.0`

2.2 GDB Path

The default GDB for this model is: `$IMPERAS_HOME/lib/$IMPERAS_ARCH/gdb/aarch64-none-elf-gdb`.

2.3 Semi-Host Library

The default semi-host library file is `arm.ovpworld.org/semihosting/armAngel/1.0`

2.4 Processor Endian-ness

This is a LITTLE endian model.

2.5 QuantumLeap Support

This processor is qualified to run in a QuantumLeap enabled simulator.

2.6 Processor ELF code

ELF codes supported by this model are: 0xb7 and 0x28.

Chapter 3

All Variants in this model

This model has these variants

Variant	Description
ARMv4T	
ARMv4xM	
ARMv4	
ARMv4TxM	
ARMv5xM	
ARMv5	
ARMv5TxM	
ARMv5T	
ARMv5TExP	
ARMv5TE	
ARMv5TEJ	
ARMv6	
ARMv6K	
ARMv6T2	
ARMv6KZ	
ARMv7	
ARM7TDMI	
ARM7EJ-S	
ARM720T	
ARM920T	
ARM922T	
ARM926EJ-S	
ARM940T	
ARM946E	
ARM966E	
ARM968E-S	
ARM1020E	
ARM1022E	
ARM1026EJ-S	
ARM1136J-S	
ARM1156T2-S	

ARM1176JZ-S	
Cortex-R4	
Cortex-R4F	
Cortex-R5MPx1	
Cortex-R5MPx2	
Cortex-R8MPx1	
Cortex-R8MPx2	
Cortex-R8MPx3	
Cortex-R8MPx4	
Cortex-R52MPx1	
Cortex-R52MPx2	
Cortex-R52MPx3	
Cortex-R52MPx4	
Cortex-R52+MPx1	
Cortex-R52+MPx2	
Cortex-R52+MPx3	
Cortex-R52+MPx4	
Cortex-R82MPx1	
Cortex-R82MPx2	
Cortex-R82MPx3	
Cortex-R82MPx4	
Cortex-R82MPx5	
Cortex-R82MPx6	
Cortex-R82MPx7	
Cortex-R82MPx8	
Cortex-A5UP	
Cortex-A5MPx1	
Cortex-A5MPx2	
Cortex-A5MPx3	
Cortex-A5MPx4	
Cortex-A8	
Cortex-A9UP	
Cortex-A9MPx1	
Cortex-A9MPx2	
Cortex-A9MPx3	
Cortex-A9MPx4	
Cortex-A7UP	
Cortex-A7MPx1	
Cortex-A7MPx2	
Cortex-A7MPx3	
Cortex-A7MPx4	
Cortex-A15UP	
Cortex-A15MPx1	
Cortex-A15MPx2	
Cortex-A15MPx3	

Cortex-A15MPx4	
Cortex-A17MPx1	
Cortex-A17MPx2	
Cortex-A17MPx3	
Cortex-A17MPx4	
AArch32	
AArch64	
Cortex-A32MPx1	
Cortex-A32MPx2	
Cortex-A32MPx3	
Cortex-A32MPx4	
Cortex-A35MPx1	
Cortex-A35MPx2	
Cortex-A35MPx3	
Cortex-A35MPx4	
Cortex-A53MPx1	
Cortex-A53MPx2	
Cortex-A53MPx3	
Cortex-A53MPx4	
Cortex-A55MPx1	
Cortex-A55MPx2	
Cortex-A55MPx3	
Cortex-A55MPx4	
Cortex-A57MPx1	
Cortex-A57MPx2	
Cortex-A57MPx3	
Cortex-A57MPx4	
Cortex-A65AEMPx1	
Cortex-A65AEMPx2	
Cortex-A65AEMPx3	
Cortex-A65AEMPx4	
Cortex-A72MPx1	
Cortex-A72MPx2	
Cortex-A72MPx3	
Cortex-A72MPx4	
Cortex-A73MPx1	
Cortex-A73MPx2	
Cortex-A73MPx3	
Cortex-A73MPx4	
Cortex-A75MPx1	
Cortex-A75MPx2	
Cortex-A75MPx3	
Cortex-A75MPx4	
Cortex-A76MPx1	
Cortex-A76MPx2	

Cortex-A76MPx3	
Cortex-A76MPx4	(described in this document)
Cortex-A78MPx1	
Cortex-A78MPx2	
Cortex-A78MPx3	
Cortex-A78MPx4	
MultiCluster	

Table 3.1: All Variants in this model

Chapter 4

Bus Master Ports

This model has these bus master ports.

Name	min	max	Connect?	Description
INSTRUCTION	32	53	mandatory	
DATA	32	53	optional	
GICRegisters	32	32	optional	GIC memory-mapped register block
GICDRegisters	32	32	optional	GICD memory-mapped register block

Table 4.1: Bus Master Ports

Chapter 5

Bus Slave Ports

This model has no bus slave ports.

Chapter 6

Net Ports

This model has these net ports.

Name	Type	Connect?	Description
SPI32	input	optional	Shared peripheral interrupt
SPI33	input	optional	Shared peripheral interrupt
SPI34	input	optional	Shared peripheral interrupt
SPI35	input	optional	Shared peripheral interrupt
SPI36	input	optional	Shared peripheral interrupt
SPI37	input	optional	Shared peripheral interrupt
SPI38	input	optional	Shared peripheral interrupt
SPI39	input	optional	Shared peripheral interrupt
SPI40	input	optional	Shared peripheral interrupt
SPI41	input	optional	Shared peripheral interrupt
SPI42	input	optional	Shared peripheral interrupt
SPI43	input	optional	Shared peripheral interrupt
SPI44	input	optional	Shared peripheral interrupt
SPI45	input	optional	Shared peripheral interrupt
SPI46	input	optional	Shared peripheral interrupt
SPI47	input	optional	Shared peripheral interrupt
SPI48	input	optional	Shared peripheral interrupt
SPI49	input	optional	Shared peripheral interrupt
SPI50	input	optional	Shared peripheral interrupt
SPI51	input	optional	Shared peripheral interrupt
SPI52	input	optional	Shared peripheral interrupt
SPI53	input	optional	Shared peripheral interrupt
SPI54	input	optional	Shared peripheral interrupt
SPI55	input	optional	Shared peripheral interrupt
SPI56	input	optional	Shared peripheral interrupt
SPI57	input	optional	Shared peripheral interrupt
SPI58	input	optional	Shared peripheral interrupt
SPI59	input	optional	Shared peripheral interrupt
SPI60	input	optional	Shared peripheral interrupt
SPI61	input	optional	Shared peripheral interrupt

SPI62	input	optional	Shared peripheral interrupt
SPI63	input	optional	Shared peripheral interrupt
SPI64	input	optional	Shared peripheral interrupt
SPI65	input	optional	Shared peripheral interrupt
SPI66	input	optional	Shared peripheral interrupt
SPI67	input	optional	Shared peripheral interrupt
SPI68	input	optional	Shared peripheral interrupt
SPI69	input	optional	Shared peripheral interrupt
SPI70	input	optional	Shared peripheral interrupt
SPI71	input	optional	Shared peripheral interrupt
SPI72	input	optional	Shared peripheral interrupt
SPI73	input	optional	Shared peripheral interrupt
SPI74	input	optional	Shared peripheral interrupt
SPI75	input	optional	Shared peripheral interrupt
SPI76	input	optional	Shared peripheral interrupt
SPI77	input	optional	Shared peripheral interrupt
SPI78	input	optional	Shared peripheral interrupt
SPI79	input	optional	Shared peripheral interrupt
SPI80	input	optional	Shared peripheral interrupt
SPI81	input	optional	Shared peripheral interrupt
SPI82	input	optional	Shared peripheral interrupt
SPI83	input	optional	Shared peripheral interrupt
SPI84	input	optional	Shared peripheral interrupt
SPI85	input	optional	Shared peripheral interrupt
SPI86	input	optional	Shared peripheral interrupt
SPI87	input	optional	Shared peripheral interrupt
SPI88	input	optional	Shared peripheral interrupt
SPI89	input	optional	Shared peripheral interrupt
SPI90	input	optional	Shared peripheral interrupt
SPI91	input	optional	Shared peripheral interrupt
SPI92	input	optional	Shared peripheral interrupt
SPI93	input	optional	Shared peripheral interrupt
SPI94	input	optional	Shared peripheral interrupt
SPI95	input	optional	Shared peripheral interrupt
SPIVector	input	optional	Shared peripheral interrupt vectorized input
periphReset	input	optional	Peripheral reset (active high)
GICCDISABLE	input	optional	GIC CPU interface logic disable (active high, sampled on rising edge of periphReset)
EVENTI	input	optional	Event input signal, active on rising edge
EVENTO	output	optional	Event output signal, active on rising edge
PPI16_CPU0	input	optional	Private peripheral interrupt
PPI17_CPU0	input	optional	Private peripheral interrupt
PPI18_CPU0	input	optional	Private peripheral interrupt

PPI19_CPU0	input	optional	Private peripheral interrupt
PPI20_CPU0	input	optional	Private peripheral interrupt
PPI21_CPU0	input	optional	Private peripheral interrupt
PPI22_CPU0	input	optional	Private peripheral interrupt
PPI23_CPU0	input	optional	Private peripheral interrupt
PPI24_CPU0	input	optional	Private peripheral interrupt
PPI25_CPU0	input	optional	Private peripheral interrupt
PPI26_CPU0	input	optional	Private peripheral interrupt
PPI27_CPU0	input	optional	Private peripheral interrupt
PPI28_CPU0	input	optional	Private peripheral interrupt
PPI29_CPU0	input	optional	Private peripheral interrupt
PPI30_CPU0	input	optional	Private peripheral interrupt
PPI31_CPU0	input	optional	Private peripheral interrupt
CNTVIRQ_CPU0	output	optional	EL1 Virtual timer event (active high)
CNTPSIRQ_CPU0	output	optional	EL3 Physical timer event (active high)
CNTPNSIRQ_CPU0	output	optional	EL1 Physical timer event (active high)
CNTPHPIRQ_CPU0	output	optional	Non-secure EL2 Physical timer event (active high)
CNTHVIRQ_CPU0	output	optional	Non-secure EL2 Virtual timer event (active high)
IRQOUT_CPU0	output	optional	IRQ wakeup
FIQOUT_CPU0	output	optional	FIQ wakeup
CLUSTERIDAFF1	input	optional	Configure MPIDR.Aff1
CLUSTERIDAFF2	input	optional	Configure MPIDR.Aff2
CLUSTERIDAFF3	input	optional	Configure MPIDR.Aff3
RVBARADDRx_CPU0	input	optional	Configure AArch64 Reset Vector Base Address at reset
VINITI1_CPU0	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU0	input	optional	Configure exception endianness (SCTLR.EE)
CFGTE_CPU0	input	optional	Configure exception state at reset (SCTLR.TE)
reset_CPU0	input	optional	Processor reset, active high
fiq_CPU0	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU0	input	optional	IRQ interrupt, active high (negation of nIRQ)
sei_CPU0	input	optional	System error interrupt, active on rising edge (negation of nSEI)
vfiq_CPU0	input	optional	Virtual FIQ interrupt, active high (negation of nVFIQ)
virq_CPU0	input	optional	Virtual IRQ interrupt, active high (negation of nVIRQ)
vsei_CPU0	input	optional	Virtual system error interrupt, active on rising edge (negation of nVSEI)

haltReason_CPU0	output	optional	Indicates why core is halted
AXI.SLVERR_CPU0	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
CP15SDISABLE_CPU0	input	optional	CP15SDISABLE (active high)
PMUIRQ_CPU0	output	optional	Performance monitor event (active high)
syncCluster_CPU0	output	optional	Cluster synchronization required
updateTimer_CPU0	output	optional	Internal timer update
PPI16_CPU1	input	optional	Private peripheral interrupt
PPI17_CPU1	input	optional	Private peripheral interrupt
PPI18_CPU1	input	optional	Private peripheral interrupt
PPI19_CPU1	input	optional	Private peripheral interrupt
PPI20_CPU1	input	optional	Private peripheral interrupt
PPI21_CPU1	input	optional	Private peripheral interrupt
PPI22_CPU1	input	optional	Private peripheral interrupt
PPI23_CPU1	input	optional	Private peripheral interrupt
PPI24_CPU1	input	optional	Private peripheral interrupt
PPI25_CPU1	input	optional	Private peripheral interrupt
PPI26_CPU1	input	optional	Private peripheral interrupt
PPI27_CPU1	input	optional	Private peripheral interrupt
PPI28_CPU1	input	optional	Private peripheral interrupt
PPI29_CPU1	input	optional	Private peripheral interrupt
PPI30_CPU1	input	optional	Private peripheral interrupt
PPI31_CPU1	input	optional	Private peripheral interrupt
CNTVIRQ_CPU1	output	optional	EL1 Virtual timer event (active high)
CNTPSIRQ_CPU1	output	optional	EL3 Physical timer event (active high)
CNTPNSIRQ_CPU1	output	optional	EL1 Physical timer event (active high)
CNTPHPIRQ_CPU1	output	optional	Non-secure EL2 Physical timer event (active high)
CNTHVIRQ_CPU1	output	optional	Non-secure EL2 Virtual timer event (active high)
IRQOUT_CPU1	output	optional	IRQ wakeup
FIQOUT_CPU1	output	optional	FIQ wakeup
RVBARADDRx_CPU1	input	optional	Configure AArch64 Reset Vector Base Address at reset
VINITHL_CPU1	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU1	input	optional	Configure exception endianness (SCTLR.EE)
CFGTE_CPU1	input	optional	Configure exception state at reset (SCTLR.TE)
reset_CPU1	input	optional	Processor reset, active high
fiq_CPU1	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU1	input	optional	IRQ interrupt, active high (negation of nIRQ)

sei_CPU1	input	optional	System error interrupt, active on rising edge (negation of nSEI)
vfirq_CPU1	input	optional	Virtual FIQ interrupt, active high (negation of nVFIQ)
virq_CPU1	input	optional	Virtual IRQ interrupt, active high (negation of nVIRQ)
vsei_CPU1	input	optional	Virtual system error interrupt, active on rising edge (negation of nVSEI)
haltReason_CPU1	output	optional	Indicates why core is halted
AXI_SLVERR_CPU1	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
CP15SDISABLE_CPU1	input	optional	CP15SDISABLE (active high)
PMUIRQ_CPU1	output	optional	Performance monitor event (active high)
syncCluster_CPU1	output	optional	Cluster synchronization required
updateTimer_CPU1	output	optional	Internal timer update
PPI16_CPU2	input	optional	Private peripheral interrupt
PPI17_CPU2	input	optional	Private peripheral interrupt
PPI18_CPU2	input	optional	Private peripheral interrupt
PPI19_CPU2	input	optional	Private peripheral interrupt
PPI20_CPU2	input	optional	Private peripheral interrupt
PPI21_CPU2	input	optional	Private peripheral interrupt
PPI22_CPU2	input	optional	Private peripheral interrupt
PPI23_CPU2	input	optional	Private peripheral interrupt
PPI24_CPU2	input	optional	Private peripheral interrupt
PPI25_CPU2	input	optional	Private peripheral interrupt
PPI26_CPU2	input	optional	Private peripheral interrupt
PPI27_CPU2	input	optional	Private peripheral interrupt
PPI28_CPU2	input	optional	Private peripheral interrupt
PPI29_CPU2	input	optional	Private peripheral interrupt
PPI30_CPU2	input	optional	Private peripheral interrupt
PPI31_CPU2	input	optional	Private peripheral interrupt
CNTVIRQ_CPU2	output	optional	EL1 Virtual timer event (active high)
CNTPSIRQ_CPU2	output	optional	EL3 Physical timer event (active high)
CNTPNSIRQ_CPU2	output	optional	EL1 Physical timer event (active high)
CNTPHPIRQ_CPU2	output	optional	Non-secure EL2 Physical timer event (active high)
CNTHVIRQ_CPU2	output	optional	Non-secure EL2 Virtual timer event (active high)
IRQOUT_CPU2	output	optional	IRQ wakeup
FIQOUT_CPU2	output	optional	FIQ wakeup
RVBARADDRx_CPU2	input	optional	Configure AArch64 Reset Vector Base Address at reset
VINITI1_CPU2	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU2	input	optional	Configure exception endianness (SCTLR.EE)

CFGTE_CPU2	input	optional	Configure exception state at reset (SCTLR.TE)
reset_CPU2	input	optional	Processor reset, active high
fiq_CPU2	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU2	input	optional	IRQ interrupt, active high (negation of nIRQ)
sei_CPU2	input	optional	System error interrupt, active on rising edge (negation of nSEI)
vfiq_CPU2	input	optional	Virtual FIQ interrupt, active high (negation of nVFIQ)
virq_CPU2	input	optional	Virtual IRQ interrupt, active high (negation of nVIRQ)
vsei_CPU2	input	optional	Virtual system error interrupt, active on rising edge (negation of nVSEI)
haltReason_CPU2	output	optional	Indicates why core is halted
AXI_SLVERR_CPU2	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
CP15SDISABLE_CPU2	input	optional	CP15SDISABLE (active high)
PMUIRQ_CPU2	output	optional	Performance monitor event (active high)
syncCluster_CPU2	output	optional	Cluster synchronization required
updateTimer_CPU2	output	optional	Internal timer update
PPI16_CPU3	input	optional	Private peripheral interrupt
PPI17_CPU3	input	optional	Private peripheral interrupt
PPI18_CPU3	input	optional	Private peripheral interrupt
PPI19_CPU3	input	optional	Private peripheral interrupt
PPI20_CPU3	input	optional	Private peripheral interrupt
PPI21_CPU3	input	optional	Private peripheral interrupt
PPI22_CPU3	input	optional	Private peripheral interrupt
PPI23_CPU3	input	optional	Private peripheral interrupt
PPI24_CPU3	input	optional	Private peripheral interrupt
PPI25_CPU3	input	optional	Private peripheral interrupt
PPI26_CPU3	input	optional	Private peripheral interrupt
PPI27_CPU3	input	optional	Private peripheral interrupt
PPI28_CPU3	input	optional	Private peripheral interrupt
PPI29_CPU3	input	optional	Private peripheral interrupt
PPI30_CPU3	input	optional	Private peripheral interrupt
PPI31_CPU3	input	optional	Private peripheral interrupt
CNTVIRQ_CPU3	output	optional	EL1 Virtual timer event (active high)
CNTPSIRQ_CPU3	output	optional	EL3 Physical timer event (active high)
CNTPNSIRQ_CPU3	output	optional	EL1 Physical timer event (active high)
CNTPHPIRQ_CPU3	output	optional	Non-secure EL2 Physical timer event (active high)
CNTHVIRQ_CPU3	output	optional	Non-secure EL2 Virtual timer event (active high)

IRQOUT_CPU3	output	optional	IRQ wakeup
FIQOUT_CPU3	output	optional	FIQ wakeup
RVBARADDRx_CPU3	input	optional	Configure AArch64 Reset Vector Base Address at reset
VINITH_CPU3	input	optional	Configure HIVECS mode (SCTLR.V)
CFGEND_CPU3	input	optional	Configure exception endianness (SCTLR.EE)
CFGTE_CPU3	input	optional	Configure exception state at reset (SCTLR.TE)
reset_CPU3	input	optional	Processor reset, active high
fiq_CPU3	input	optional	FIQ interrupt, active high (negation of nFIQ)
irq_CPU3	input	optional	IRQ interrupt, active high (negation of nIRQ)
sei_CPU3	input	optional	System error interrupt, active on rising edge (negation of nSEI)
vfiq_CPU3	input	optional	Virtual FIQ interrupt, active high (negation of nVFIQ)
virq_CPU3	input	optional	Virtual IRQ interrupt, active high (negation of nVIRQ)
vsei_CPU3	input	optional	Virtual system error interrupt, active on rising edge (negation of nVSEI)
haltReason_CPU3	output	optional	Indicates why core is halted
AXI_SLVERR_CPU3	input	optional	AXI external abort type (DECERR=0, SLVERR=1)
CP15SDISABLE_CPU3	input	optional	CP15SDISABLE (active high)
PMUIRQ_CPU3	output	optional	Performance monitor event (active high)
syncCluster_CPU3	output	optional	Cluster synchronization required
updateTimer_CPU3	output	optional	Internal timer update

Table 6.1: Net Ports

Chapter 7

FIFO Ports

This model has no FIFO ports.

Chapter 8

Formal Parameters

Name	Type	Description
verbose	Boolean	Specify verbosity of output
suppressCPSWarnings	Boolean	Suppress duplicate warnings generated using ARM_CP_CPSI or ARM_CP_CPSD message identifiers
showHiddenRegs	Boolean	Show hidden registers during register tracing
UAL	Boolean	Disassemble using UAL syntax
disableGICModel	Boolean	Disable the internal GIC model entirely
enableGICv3	Boolean	Enable/disable GICv3 support
enableGICv2_64kB_Page	Boolean	Enable 64kB page size for GICv2 memory-mapped register groups (Xilinx Zynq Ultra-scale support)
supportSTATUSR	Boolean	Enable/disable support for GICv3 GIC[CDV]_STATUSR registers
enableVFPAAtReset	Boolean	Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)
SVEImplementedSizes	Uns32	For processors with ARMv8.2 SVE extension, mask of configurable vector sizes (vector length N is configurable if mask contains $1 < ((N/128)-1)$)
SVEFaultUnknown	Uns64	For processors with ARMv8.2 SVE extension, UNKNOWN value returned for suppressed or inactive FFR elements
enableSystemBus	Boolean	Add 32-bit artifact System bus port, allowing system registers to be externally implemented
enableSystemMonitorBus	Boolean	Add 32-bit artifact SystemMonitor bus port, allowing system register accesses to be externally monitored
distinctMTCores	Boolean	For multi-threaded (MT) processors, simulate threads as separate cores (otherwise, simulate MT threads as a single entity)
compatibility	Enumeration	Specify compatibility mode
	ISA	
	gdb	
	nopSVC	
unpredictableR15	Enumeration	Specify behavior for UNPREDICTABLE uses of AArch32 R15 register
	undefined	
	nop	
	raz_wi	

	execute	
	assert	
unpredictableModal	Enumeration	Specify behavior for UNPREDICTABLE instructions in certain AArch32 modes (for example, MRS using SPSR in System mode)
	undefined	
	nop	
	assert	
maxSIMDUnroll	Uns32	If SIMD operations are supported, specify the maximum number of parallel SIMD operations to unroll (unrolled operations can be faster, but produce more verbose JIT code)
override_debugMask	Uns32	Specifies debug mask, enabling debug output for model components
ASIDCacheSize	Uns32	Specifies the number of different ASIDs for which TLB entries are cached; a value of 0 implies no limit
thumbNoCond	Boolean	Specify whether trapped Thumb instructions set CV=1 and COND field in syndrome (if False, both are zero)
enableHostAtomics	Boolean	Enable use of host atomic instructions to emulate load/store exclusive operations (not architecturally accurate, accelerates parallel simulation)
enableWFEHostAtomics	Boolean	Enable WFE and SEV instructions when using host atomics (does nothing otherwise)
override_numCPUs	Uns32	Specify the number of cores in a multiprocessor (maximum of 8 for GICv1/GICv2)
override_affinityMask	Uns32	Specify bitmask of implemented affinity bits in format Aff3:Aff2:Aff1:Aff0 (each a byte)
override_MPIDR_MT	Boolean	Specifies that processor is multithreaded
override_MPIDR_Aff0	Uns32	Override Aff0 field in MPIDR/MPIDR_EL1 register
override_MPIDR_Aff1	Uns32	Override Aff1 field in MPIDR/MPIDR_EL1 register (also possible by writing CLUSTERIDAFF1 configuration net)
override_MPIDR_Aff2	Uns32	Override Aff2 field in MPIDR/MPIDR_EL1 register (also possible by writing CLUSTERIDAFF2 configuration net)
override_MPIDR_Aff3	Uns32	Override Aff3 field in MPIDR_EL1 register (also possible by writing CLUSTERIDAFF3 configuration net)
override_fcsePresent	Boolean	Specifies that FCSE is present (if true)
override_fpexcDexPresent	Boolean	Specifies that the FPEXC.DEX register field is implemented (if true)
override_advSIMDPresent	Boolean	Specifies that Advanced SIMD extensions are present (if true)
override_vfpPresent	Boolean	Specifies that VFP extensions are present (if true)
override_physicalBits	Uns32	Specifies the implemented physical bus bits (defaults to connected physical bus width)
override_timerScaleFactor	Uns32	Specifies the fraction of MIPS rate to use for MPCore timers (generic timers or global/local/watchdogs depending on implementation). Defaults to 20 for generic timers, 2 for others

override_GICD_NSACRPresent	Boolean	Specifies that optional GICD_NSACR distributor registers are present (GICv2 only)
override_GICD_PPISRPresent	Boolean	Specifies that implementation-specific GICD_PPISR distributor register is present (GICv1 ICDPPIS/ICPPISR, GICv1 and GICv2 only)
override_GICD_SPISRPresent	Boolean	Specifies that implementation-specific GICD_SPISR distributor registers are present (GICv1 ICDSPIS/ICSPISR)
override_GICv3_DistributorBase	Uns64	Specify distributor register block base address (GICv3 only)
override_GICv3_E1NWFPresent	Boolean	Specifies that GICR_CTLR.E1NWF is implemented (GICv3 only)
override_GICv3_ROAffinity	Boolean	Specifies that affinity bits in the GICD_IROUTER<n> registers are initialised from MPIDR and Read-Only
override_GIC_PPIMask	Uns32	Specify bitmask of implemented PPIs in the GIC (e.g. ID16 is 0x0001, ID31 is 0x8000)
override_GICCDISABLE	Boolean	Specify initial value of GICCDISABLE
allowMSRToCurrentBank	Boolean	Allow access to the current mode registers via the MSR/MRS (Banked Register) instructions
override_SCTLR_V	Boolean	Override SCTLR.V with the passed value (enables high vectors; also configurable using VINITHI pin)
override_SCTLR_EE	Boolean	Override SCTLR.EE with the passed value (configures exception data endianness; also configurable using CFGEE pin)
override_SCTLR_TE	Boolean	Override SCTLR.TE with the passed value (configures Thumb state for exception handling; also configurable using TEINIT pin)
override_SCTLR_NMFI	Boolean	Override SCTLR.NMFI with the passed value (configures NMFI state for exception handling; also configurable using CFGNMFI pin)
override_SCTLR_CP15BEN_Present	Boolean	Enable ARMv7 SCTLR.CP15BEN bit (CP15 barrier enable)
override_MIDR	Uns32	Override MIDR/MIDR_EL1 register
override_CTR	Uns32	Override CTR/CTR_EL0 register
override_TLBTR	Uns32	Override TLBTR register
override_CLIDR	Uns32	Override CLIDR/CLIDR_EL1 register
override_ACTLR_SMP	Boolean	Override ACTLR SMP enable bit
override_AIDR	Uns32	Override AIDR/AIDR_EL1 register
override_CBAR	Uns32	Override Configuration Base Address Register (corresponds to value on PERIPHBASE configuration input)
override_PFR0	Uns32	Override ID_PFR0/ID_PFR0_EL1 register
override_PFR1	Uns32	Override ID_PFR1/ID_PFR1_EL1 register
override_PFR2	Uns32	Override ID_PFR2/ID_PFR2_EL1 register
override_DFR0	Uns32	Override ID_DFR0/ID_DFR0_EL1 register
override_DFR1	Uns32	Override ID_DFR1/ID_DFR1_EL1 register
override_AFR0	Uns32	Override ID_AFR0/ID_AFR0_EL1 register
override_MMFR0	Uns32	Override ID_MMFR0/ID_MMFR0_EL1 register
override_MMFR1	Uns32	Override ID_MMFR1/ID_MMFR1_EL1 register
override_MMFR2	Uns32	Override ID_MMFR2/ID_MMFR2_EL1 register

override_MMFR3	Uns32	Override ID_MMFR3/ID_MMFR3_EL1 register
override_MMFR4	Uns32	Override ID_MMFR4/ID_MMFR4_EL1 register
override_MMFR5	Uns32	Override ID_MMFR5/ID_MMFR5_EL1 register
override_ISAR0	Uns32	Override ID_ISAR0/ID_ISAR0_EL1 register
override_ISAR1	Uns32	Override ID_ISAR1/ID_ISAR1_EL1 register
override_ISAR2	Uns32	Override ID_ISAR2/ID_ISAR2_EL1 register
override_ISAR3	Uns32	Override ID_ISAR3/ID_ISAR3_EL1 register
override_ISAR4	Uns32	Override ID_ISAR4/ID_ISAR4_EL1 register
override_ISAR5	Uns32	Override ID_ISAR5/ID_ISAR5_EL1 register
override_ISAR6	Uns32	Override ID_ISAR6/ID_ISAR6_EL1 register
override_PMCRR	Uns32	Override PMCR/PMCR_EL0 register (not functionally significant in the model)
override_PMCEID0	Uns64	Override PMCEID0/PMCEID0_EL0 register (not functionally significant in the model)
override_PMCEID1	Uns64	Override PMCEID1/PMCEID1_EL0 register (not functionally significant in the model)
override_PMMIR	Uns32	Override PMMIR/PMMIR_EL1 register (not functionally significant in the model)
override_DBGDIDR	Uns32	Override DBGDIDR register (not functionally significant in the model)
override_DBGDEVID	Uns32	Override DBGDEVID register (not functionally significant in the model)
override_DBGDEVID1	Uns32	Override DBGDEVID1 register (not functionally significant in the model)
override_DBGDEVID2	Uns32	Override DBGDEVID2 register (not functionally significant in the model)
override_FPSID	Uns32	Override SIMD/VFP FPSID register
override_MVFR0	Uns32	Override SIMD/VFP MVFR0/MVFR0_EL1 register
override_MVFR1	Uns32	Override SIMD/VFP MVFR1/MVFR1_EL1 register
override_MVFR2	Uns32	Override SIMD/VFP MVFR2/MVFR2_EL1 register
override_FPEXC	Uns32	Override SIMD/VFP FPEXC/FPEXC32_EL2 register
override_GICC_IIDR	Uns32	Override GICC_IIDR register (GICv1 ICCI-IDR)
override_GICD_TYPER	Uns32	Override GICD_TYPER register (GICv1 IC-DICTR)
override_GICD_TYPER_ITLines	Uns32	Override ITLinesNumber field of GICD_TYPER register (GICv1 ICDICTR)
override_GICD_TYPER_ESPI	Boolean	Override ESPI field of GICD_TYPER register (GICv3.1 and later)
override_GICD_TYPER_ESPI_range	Uns32	Override ESPI_range field of GICD_TYPER register (GICv3.1 and later)
override_GICD_ICFGRN	Uns32	Override reset value of GICD_ICFGR2...GICD_ICFGRn (GICv1 ICDICFR2...ICDICFRn)
override_GICD_IIDR	Uns32	Override GICD_IIDR register (GICv1 ICDI-IDR)
override_GICH_VTR	Uns32	Override GICH_VTR register
override_GICR_IIDR	Uns32	Override GICR_IIDR register (GICv3 and later)

override_GICR_TYPER	Uns64	Override GICR_TYPER register (GICv3 and later)
override_GICR_TYPER_PPInum	Uns32	Override PPInum field of GICR_TYPER register (GICv3.1 and later)
override_GITS_IIDR	Uns32	Override GITS_IIDR register (GICv3 and later)
override_GITS_TYPER	Uns64	Override GITS_TYPER register (GICv3 and later)
override_ICCPMRBits	Uns32	Specify the number of writable bits in GICC_PMR (GICv1 ICCPMR)
override_minICCBPR	Uns32	Specify the minimum possible value for GICC_BPR (GICv1 ICCBPR)
override_ERG	Uns32	Specifies exclusive reservation granule
override_CCSIDR_1I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 1 instruction)
override_CCSIDR_1D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 1 data)
override_CCSIDR_2I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 2 instruction)
override_CCSIDR_2D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 2 data)
override_CCSIDR_3I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 3 instruction)
override_CCSIDR_3D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 3 data)
override_CCSIDR_4I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 4 instruction)
override_CCSIDR_4D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 4 data)
override_CCSIDR_5I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 5 instruction)
override_CCSIDR_5D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 5 data)
override_CCSIDR_6I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 6 instruction)
override_CCSIDR_6D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 6 data)
override_CCSIDR_7I	Uns64	Override CCSIDR/CCSIDR_EL1 (level 7 instruction)
override_CCSIDR_7D	Uns64	Override CCSIDR/CCSIDR_EL1 (level 7 data)
override_RMR	Uns32	Override RMR register alias at highest-implemented exception level
override_RVBAR	Uns64	Override RVBAR register alias at highest-implemented exception level
override_AA64PFR0_EL1	Uns64	Override ID_AA64PFR0_EL1 register
override_AA64PFR1_EL1	Uns64	Override ID_AA64PFR1_EL1 register
override_AA64DFR0_EL1	Uns64	Override ID_AA64DFR0_EL1 register
override_AA64DFR1_EL1	Uns64	Override ID_AA64DFR1_EL1 register
override_AA64AFR0_EL1	Uns64	Override ID_AA64AFR0_EL1 register
override_AA64AFR1_EL1	Uns64	Override ID_AA64AFR1_EL1 register
override_AA64ISAR0_EL1	Uns64	Override ID_AA64ISAR0_EL1 register
override_AA64ISAR1_EL1	Uns64	Override ID_AA64ISAR1_EL1 register
override_AA64ISAR2_EL1	Uns64	Override ID_AA64ISAR2_EL1 register
override_AA64MMFR0_EL1	Uns64	Override ID_AA64MMFR0_EL1 register
override_AA64MMFR1_EL1	Uns64	Override ID_AA64MMFR1_EL1 register

override_AA64MMFR2_EL1	Uns64	Override ID_AA64MMFR2_EL1 register
override_DCZID_EL0	Uns32	Override DCZID_EL0 register
override_LORID_EL1	Uns32	Override LORID_EL1 register (ARMv8.1 only)
override_STROffsetPC12	Boolean	Specifies that STR/STR of PC should do so with 12:byte offset from the current instruction (if true), otherwise an 8:byte offset is used
override_fcseRequiresMMU	Boolean	Specifies that FCSE is active only when MMU is enabled (if true)
override_ignoreBadCp15	Boolean	Specifies whether invalid coprocessor 15 access should be ignored (if true) or cause Invalid Instruction exceptions (if false)
override_SGIDisable	Boolean	Override whether GIC SGIs may be disabled (if true) or are permanently enabled (if false)
override_condUndefined	Boolean	Force undefined instructions to take Undefined Instruction exception even if they are conditional
override_deviceStrongAligned	Boolean	Force accesses to Device and Strongly Ordered regions to be aligned
override_stage1SZMinFault	Boolean	Enable Level 0 Translation faults when stage 1 TCR_ELx.TxSZ <minimum (by default, clamp to minimum)
override_stage1SZMaxFault	Boolean	Enable Level 0 Translation faults when stage 1 TCR_ELx.TxSZ >maximum (by default, clamp to maximum)
override_stage2SZMinFault	Boolean	Enable Level 0 Translation faults when stage 2 VTCR_EL2.T0SZ <minimum (by default, clamp to minimum)
override_stage2SZMaxFault	Boolean	Enable Level 0 Translation faults when stage 2 VTCR_EL2.T0SZ >maximum (by default, clamp to maximum)
override_mask_ACTLR_EL1	Uns64	Override mask of writable bits in AArch64 ACTLR_EL1 register, or AArch32 non-secure ACTLR/ACTLR2 pair, if implemented
override_mask_ACTLR_EL2	Uns64	Override mask of writable bits in AArch64 ACTLR_EL2 register, or AArch32 HACTLR/HACTLR2 pair, if implemented
override_mask_ACTLR_EL3	Uns64	Override mask of writable bits in AArch64 ACTLR_EL3 register, or AArch32 secure ACTLR/ACTLR2 pair, if implemented
override_Control_V	Boolean	Override SCTLR.V with the passed value (deprecated, use override_SCTLR_V)
override_MainId	Uns32	Override MIDR register (deprecated, use override_MIDR)
override_CacheType	Uns32	Override CTR register (deprecated, use override_CTR)
override_TLBType	Uns32	Override TLBTR register (deprecated, use override_TLBTR)
override_InstructionAttributes0	Uns32	Override ID_ISAR0 register (deprecated, use override_ISAR0)
override_InstructionAttributes1	Uns32	Override ID_ISAR1 register (deprecated, use override_ISAR1)
override_InstructionAttributes2	Uns32	Override ID_ISAR2 register (deprecated, use override_ISAR2)
override_InstructionAttributes3	Uns32	Override ID_ISAR3 register (deprecated, use override_ISAR3)

override_InstructionAttributes4	Uns32	Override ID_ISAR4 register (deprecated, use override.ISAR4)
override_InstructionAttributes5	Uns32	Override ID_ISAR5 register (deprecated, use override.ISAR5)

Table 8.1: Parameters that can be set in: MPCORE

8.1 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
(Others)				
variant				Cortex-A76MPx4
verbose			t	t
suppressCPSWarnings			f	f
showHiddenRegs			f	f
UAL			t	t
disableGICModel			f	f
enableGICv3			t	t
enableGICv2_64kB_Page			f	f
supportSTATUSR			t	t
enableVFPAAtReset			f	f
SVEImplementedSizes	1	0xffff	15	15
SVEFaultUnknown	0	0xffffffffffffff	0xdfdfdfdfdfdfdfdf	0xdfdfdfdfdfdfdfdf
enableSystemBus			f	f
enableSystemMonitorBus			f	f
distinctMTCores			f	f
compatibility			ISA	ISA
unpredictableR15			undefined	undefined
unpredictableModal			undefined	undefined
maxSIMDUnroll	1	16	2	2
override_debugMask	0	0xffffffff	0	0
ASIDCacheSize	0	0x100	8	8
thumbNoCond			f	f
enableHostAtomics			f	f
enableWFEHostAtomics			f	f
endian			none	none
override_numCPUs	0	32	4	4
override_affinityMask	0	0xffffffff	0xffff0700	0xffff0700
override_MPIDR_MT			t	t
override_MPIDR_Aff0	0	255	0	0
override_MPIDR_Aff1	0	255	0	0
override_MPIDR_Aff2	0	255	0	0
override_MPIDR_Aff3	0	255	0	0
override_fcsePresent			f	f

override_fpexcDexPresent			t	t
override_advSIMDPresent			f	f
override_vfpPresent			f	f
override_physicalBits	32	52	32	32
override_timerScaleFactor	1	0x1ff	20	20
override_GICD_NSACRPresent			f	f
override_GICD_PPISRPresent			t	t
override_GICD_SPISRPresent			t	t
override_GICv3_DistributorBase	0	0xffffffffffff	0x2f000000	0x2f000000
override_GICv3_E1NWFPresent			f	f
override_GICv3_ROAffinity			f	f
override_GIC_PPIMask	0	0xffff	0	0
override_GICCDISABLE			f	f
allowMSRToCurrentBank			f	f
override_SCTLR_V			f	f
override_SCTLR_EE			f	f
override_SCTLR_TE			f	f
override_SCTLR_NMFI			f	f
override_SCTLR_CP15BEN_Present			t	t
override_MIDR	0	0xffffffff	0x414fd0b1	0x414fd0b1
override_CTR	0	0xffffffff	0x8444c004	0x8444c004
override_TLBTR	0	0xffffffff	0	0
override_CLIDR	0	0xffffffff	0xc3000123	0xc3000123
override_ACTLR_SMP			f	f
override_AIDR	0	0xffffffff	0	0
override_CBAR	0	0xffffffff	0	0
override_PFR0	0	0xffffffff	0x10010131	0x10010131
override_PFR1	0	0xffffffff	0x10011011	0x10011011
override_PFR2	0	0xffffffff	17	17
override_DFR0	0	0xffffffff	0x4010088	0x4010088
override_DFR1	0	0xffffffff	0	0
override_AFR0	0	0xffffffff	0	0
override_MMFR0	0	0xffffffff	0x10201105	0x10201105
override_MMFR1	0	0xffffffff	0x40000000	0x40000000
override_MMFR2	0	0xffffffff	0x1260000	0x1260000
override_MMFR3	0	0xffffffff	0x2122211	0x2122211
override_MMFR4	0	0xffffffff	0x21110	0x21110
override_MMFR5	0	0xffffffff	0	0
override_ISAR0	0	0xffffffff	0x2101110	0x2101110
override_ISAR1	0	0xffffffff	0x13112111	0x13112111
override_ISAR2	0	0xffffffff	0x21232042	0x21232042
override_ISAR3	0	0xffffffff	0x1112131	0x1112131
override_ISAR4	0	0xffffffff	0x10142	0x10142
override_ISAR5	0	0xffffffff	0x1011121	0x1011121
override_ISAR6	0	0xffffffff	16	16

override_PMCR	0	0xffffffff	0x414a3000	0x414a3000
override_PMCEID0	0	0xffffffffffffff	0xfbff7f3f	0xfbff7f3f
override_PMCEID1	0	0xffffffffffffff	0x1f1ae7b	0x1f1ae7b
override_PMMIR	0	0xffffffff	0	0
override_DBGDIDR	0	0xffffffff	0x3518c000	0x3518c000
override_DBGDEVID	0	0xffffffff	0x100f10	0x100f10
override_DBGDEVID1	0	0xffffffff	0	0
override_DBGDEVID2	0	0xffffffff	0	0
override_FPSID	0	0xffffffff	0x41034080	0x41034080
override_MVFR0	0	0xffffffff	0x10110222	0x10110222
override_MVFR1	0	0xffffffff	0x13211111	0x13211111
override_MVFR2	0	0xffffffff	67	67
override_FPEXC	0	0xffffffff	0	0
override_GICC_IIDR	0	0xffffffff	0x4043b	0x4043b
override_GICD_TYPER	0	0xffffffff	0x7bfc02	0x7bfc02
override_GICD_TYPER_ITLines	0	31	2	2
override_GICD_TYPER_ESPI			f	f
override_GICD_TYPER_ESPI_range	0	31	0	0
override_GICD_ICFGRN	0	0xffffffff	0	0
override_GICD_IIDR	0	0xffffffff	0x102043b	0x102043b
override_GICH_VTR	0	0xffffffff	0x90000003	0x90000003
override_GICR_IIDR	0	0xffffffff	0x43b	0x43b
override_GICR_TYPER	0	0xffffffffffffff	9	9
override_GICR_TYPER_PPInum	0	2	0	0
override_GITS_IIDR	0	0xffffffff	0x43b	0x43b
override_GITS_TYPER	0	0xffffffffffffff	0x9ef79	0x9ef79
override_ICCPMRBits	4	8	5	5
override_minICCBPR	0	7	2	2
override_ERG	3	11	4	4
override_CCSIDR_1I	0	0xffffffffffffff	0x701fe01a	0x701fe01a
override_CCSIDR_1D	0	0xffffffffffffff	0x201fe01a	0x201fe01a
override_CCSIDR_2I	0	0xffffffffffffff	0x703fe03a	0x703fe03a
override_CCSIDR_2D	0	0xffffffffffffff	0x703fe03a	0x703fe03a
override_CCSIDR_3I	0	0xffffffffffffff	0x73ffe07a	0x73ffe07a
override_CCSIDR_3D	0	0xffffffffffffff	0x73ffe07a	0x73ffe07a
override_CCSIDR_4I	0	0xffffffffffffff	0	0
override_CCSIDR_4D	0	0xffffffffffffff	0	0
override_CCSIDR_5I	0	0xffffffffffffff	0	0
override_CCSIDR_5D	0	0xffffffffffffff	0	0
override_CCSIDR_6I	0	0xffffffffffffff	0	0
override_CCSIDR_6D	0	0xffffffffffffff	0	0
override_CCSIDR_7I	0	0xffffffffffffff	0	0
override_CCSIDR_7D	0	0xffffffffffffff	0	0
override_RMR	0	0xffffffff	1	1
override_RVBAR	0	0xffffffffffffff	0	0

override_AA64PFR0_EL1	0	0xffffffffffff	0x11111112	0x11111112
override_AA64PFR1_EL1	0	0xffffffffffff	16	16
override_AA64DFR0_EL1	0	0xffffffffffff	0x10305408	0x10305408
override_AA64DFR1_EL1	0	0xffffffffffff	0	0
override_AA64AFR0_EL1	0	0xffffffffffff	0	0
override_AA64AFR1_EL1	0	0xffffffffffff	0	0
override_AA64ISAR0_EL1	0	0xffffffffffff	0x10211120	0x10211120
override_AA64ISAR1_EL1	0	0xffffffffffff	0x100001	0x100001
override_AA64ISAR2_EL1	0	0xffffffffffff	0	0
override_AA64MMFR0_EL1	0	0xffffffffffff	0x101122	0x101122
override_AA64MMFR1_EL1	0	0xffffffffffff	0x10212122	0x10212122
override_AA64MMFR2_EL1	0	0xffffffffffff	0x1011	0x1011
override_DCZID_EL0	0	9	4	4
override_LORID_EL1	0	0xffffffff	0x40004	0x40004
override_STROffsetPC12			t	t
override_fcseRequiresMMU			f	f
override_ignoreBadCp15			f	f
override_SGIDisable			f	f
override_condUndefined			f	f
override_deviceStrongAligned			t	t
override_stage1SZMinFault			f	f
override_stage1SZMaxFault			f	f
override_stage2SZMinFault			f	f
override_stage2SZMaxFault			f	f
override_mask_ACTLR_EL1	0	0xffffffffffff	0	0
override_mask_ACTLR_EL2	0	0xffffffffffff	0x1cb3	0x1cb3
override_mask_ACTLR_EL3	0	0xffffffffffff	0x1cb3	0x1cb3
override_Control_V			f	f
override_MainId	0	0xffffffff	0x414fd0b1	0x414fd0b1
override_CacheType	0	0xffffffff	0x8444c004	0x8444c004
override_TLBType	0	0xffffffff	0	0
override_InstructionAttributes0	0	0xffffffff	0x2101110	0x2101110
override_InstructionAttributes1	0	0xffffffff	0x13112111	0x13112111
override_InstructionAttributes2	0	0xffffffff	0x21232042	0x21232042
override_InstructionAttributes3	0	0xffffffff	0x1112131	0x1112131
override_InstructionAttributes4	0	0xffffffff	0x10142	0x10142
override_InstructionAttributes5	0	0xffffffff	0x1011121	0x1011121

Table 8.2: Parameter values and limits

Name	Type	Description
enableVFPAAtReset	Boolean	Enable vector floating point (SIMD and VFP) instructions at reset. (Enables cp10/11 in CPACR and sets FPEXC.EN)
override_advSIMDPresent	Boolean	Specifies that Advanced SIMD extensions are present (if true)

override_vfpPresent	Boolean	Specifies that VFP extensions are present (if true)
override_SCTLR_V	Boolean	Override SCTLR.V with the passed value (enables high vectors; also configurable using VINITHI pin)
override_SCTLR_EE	Boolean	Override SCTLR.EE with the passed value (configures exception data endianness; also configurable using CFGEE pin)
override_SCTLR_TE	Boolean	Override SCTLR.TE with the passed value (configures Thumb state for exception handling; also configurable using TEINIT pin)
override_SCTLR_NMFI	Boolean	Override SCTLR.NMFI with the passed value (configures NMFI state for exception handling; also configurable using CFGNMFI pin)
override_SCTLR_CP15BEN_Present	Boolean	Enable ARMv7 SCTLR.CP15BEN bit (CP15 barrier enable)
override_ACTLR_SMP	Boolean	Override ACTLR SMP enable bit
override_RVBAR	Uns64	Override RVBAR register alias at highest-implemented exception level

Table 8.3: Parameters that can be set in: CPU

8.2 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
(Others)				
enableVFPAAtReset			f	f
override_advSIMDPresent			f	f
override_vfpPresent			f	f
override_SCTLR_V			f	f
override_SCTLR_EE			f	f
override_SCTLR_TE			f	f
override_SCTLR_NMFI			f	f
override_SCTLR_CP15BEN_Present			t	t
override_ACTLR_SMP			f	f
override_RVBAR	0	0xffffffffffffff	0	0

Table 8.4: Parameter values and limits

Chapter 9

Execution Modes

Mode	Code
EL0t	0
EL1t	4
EL1h	5
EL2t	8
EL2h	9
EL3t	12
EL3h	13
User	16

Table 9.1: Modes implemented in: CPU

Chapter 10

Exceptions

Exception	Code
Reset	0
Undefined	1
SupervisorCall	2
SecureMonitorCall	3
HypervisorCall	4
PrefetchAbort	5
DataAbort	6
HypervisorTrap	7
IRQ	8
FIQ	9
IllegalState	10
MisalignedPC	11
MisalignedSP	12
SError	13

Table 10.1: Exceptions implemented in: CPU

Chapter 11

Hierarchy of the model

A CPU core may be configured to instance many processors of a Symmetrical Multi Processor (SMP). A CPU core may also have sub elements within a processor, for example hardware threading blocks.

OVP processor models can be written to include SMP blocks and to have many levels of hierarchy. Some OVP CPU models may have a fixed hierarchy, and some may be configured by settings in a configuration register. Please see the register definitions of this model.

This model documentation shows the settings and hierarchy of the default settings for this model variant.

11.1 Level 1: MPCORE

This level in the model hierarchy has 2 commands.

This level in the model hierarchy has no register groups.

This level in the model hierarchy has 4 children:

CPU0, CPU1, CPU2 and CPU3.

11.2 Level 2: CPU

This level in the model hierarchy has 7 commands.

This level in the model hierarchy has 31 register groups:

Group name	Registers
Core	15
Core_AAArch64	33
Control	3
User	7
FIQ	8
IRQ	3
Supervisor	3
Monitor	3
Hypervisor	3
Undefined	3
Abort	3

SIMD_VFP	32
SIMD_VFP_SYS	6
SIMD_FP_AArch64	32
AArch32_32_bit_system	46
AArch32_32_bit_secure_system	4
AArch32_32_bit_non_secure_system	4
AArch32_64_bit_system	7
AArch32_64_bit_secure_system	1
AArch32_64_bit_non_secure_system	1
AArch64_system	346
AArch64_secure_system	19
AArch64_non_secure_system	19
AArch64_SYS_instruction_registers	58
Integration_support	37
MPCore_distributor	146
MPCore_physical_redistributor	43
MPCore_processor_interface	16
MPCore_virtual_interface_control	11
MPCore_virtual_processor_interface	31
MPCore_ITS	14

Table 11.1: Register groups

This level in the model hierarchy has no children.

Chapter 12

Model Commands

A Processor model can implement one or more **Model Commands** available to be invoked from the simulator command line, from the OP API or from the Imperas Multiprocessor Debugger.

12.1 Level 1: MPCORE

12.1.1 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addresshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.1: isync command arguments

12.1.2 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing
-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing
-processorname	Boolean	Include processor name in all trace lines

-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.2: itrace command arguments

12.2 Level 2: CPU

12.2.1 debugflags

show or modify the processor debug flags

Argument	Type	Description
-get	Boolean	print current processor flags value
-mask	Boolean	print valid debug flag bits
-set	Int32	new processor flags (only flags 0x000003e4 can be modified)

Table 12.3: debugflags command arguments

12.2.2 dumpTLB

report TLB contents

Argument	Type	Description
-all	Boolean	show the contents of all TLBs (if False, show just the current TLB)

Table 12.4: dumpTLB command arguments

12.2.3 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addressshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.5: isync command arguments

12.2.4 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing

-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing
-processorname	Boolean	Include processor name in all trace lines
-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.6: itrace command arguments

12.2.5 listSysRegsAA32

12.2.5.1 Argument description

List all AArch32 system registers

12.2.6 listSysRegsAA64

12.2.6.1 Argument description

List all AArch64 system registers

12.2.7 validateTLB

check TLB contents against page tables in memory and report incoherent entries

Argument	Type	Description
-all	Boolean	check all TLBs (if False, validate just the current TLB)
-verbose	Boolean	show all TLB entries (if False, show only incoherent entries)

Table 12.7: validateTLB command arguments

Chapter 13

Registers

13.1 Level 1: MPCORE

No registers.

13.2 Level 2: CPU

13.2.1 Core

Registers at level:2, type:CPU group:Core

Name	Bits	Initial-Hex	RW	Description
r0	32	0	rw	
r1	32	0	rw	
r2	32	0	rw	
r3	32	0	rw	
r4	32	0	rw	
r5	32	0	rw	
r6	32	0	rw	
r7	32	0	rw	
r8	32	0	rw	
r9	32	0	rw	
r10	32	0	rw	
r11	32	0	rw	frame pointer
r12	32	0	rw	
sp	32	0	rw	stack pointer
lr	32	0	rw	

Table 13.1: Registers at level 2, type:CPU group:Core

13.2.2 Core_AAArch64

Registers at level:2, type:CPU group:Core_AAArch64

Name	Bits	Initial-Hex	RW	Description
x0	64	0	rw	
x1	64	0	rw	
x2	64	0	rw	
x3	64	0	rw	
x4	64	0	rw	

x5	64	0	rw	
x6	64	0	rw	
x7	64	0	rw	
x8	64	0	rw	
x9	64	0	rw	
x10	64	0	rw	
x11	64	0	rw	
x12	64	0	rw	
x13	64	0	rw	
x14	64	0	rw	
x15	64	0	rw	
x16	64	0	rw	
x17	64	0	rw	
x18	64	0	rw	
x19	64	0	rw	
x20	64	0	rw	
x21	64	0	rw	
x22	64	0	rw	
x23	64	0	rw	
x24	64	0	rw	
x25	64	0	rw	
x26	64	0	rw	
x27	64	0	rw	
x28	64	0	rw	
x29	64	0	rw	frame pointer
x30	64	0	rw	
sp	64	0	rw	stack pointer
pc	64	0	rw	program counter

Table 13.2: Registers at level 2, type:CPU group:Core_AAArch64

13.2.3 Control

Registers at level:2, type:CPU group:Control

Name	Bits	Initial-Hex	RW	Description
fps	32	0	rw	archaic FPSCR view (for gdb)
cpsr	32	3cd	rw	
spsr	32	0	rw	

Table 13.3: Registers at level 2, type:CPU group:Control

13.2.4 User

Registers at level:2, type:CPU group:User

Name	Bits	Initial-Hex	RW	Description
r8_usr	32	0	rw	
r9_usr	32	0	rw	
r10_usr	32	0	rw	
r11_usr	32	0	rw	
r12_usr	32	0	rw	
sp_usr	32	0	rw	
lr_usr	32	0	rw	

Table 13.4: Registers at level 2, type:CPU group:User

13.2.5 FIQ

Registers at level:2, type:CPU group:FIQ

Name	Bits	Initial-Hex	RW	Description
r8_fiq	32	0	rw	
r9_fiq	32	0	rw	
r10_fiq	32	0	rw	
r11_fiq	32	0	rw	
r12_fiq	32	0	rw	
sp_fiq	32	0	rw	
lr_fiq	32	0	rw	
spsr_fiq	32	0	rw	

Table 13.5: Registers at level 2, type:CPU group:FIQ

13.2.6 IRQ

Registers at level:2, type:CPU group:IRQ

Name	Bits	Initial-Hex	RW	Description
sp_irq	32	0	rw	
lr_irq	32	0	rw	
spsr_irq	32	0	rw	

Table 13.6: Registers at level 2, type:CPU group:IRQ

13.2.7 Supervisor

Registers at level:2, type:CPU group:Supervisor

Name	Bits	Initial-Hex	RW	Description
sp_svc	32	0	rw	
lr_svc	32	0	rw	
spsr_svc	32	0	rw	

Table 13.7: Registers at level 2, type:CPU group:Supervisor

13.2.8 Monitor

Registers at level:2, type:CPU group:Monitor

Name	Bits	Initial-Hex	RW	Description
sp_mon	32	0	rw	
lr_mon	32	0	rw	
spsr_mon	32	0	rw	

Table 13.8: Registers at level 2, type:CPU group:Monitor

13.2.9 Hypervisor

Registers at level:2, type:CPU group:Hypervisor

Name	Bits	Initial-Hex	RW	Description
sp_hyp	32	0	rw	
elr_hyp	32	0	rw	
spsr_hyp	32	0	rw	

Table 13.9: Registers at level 2, type:CPU group:Hypervisor

13.2.10 Undefined

Registers at level:2, type:CPU group:Undefined

Name	Bits	Initial-Hex	RW	Description
sp_undef	32	0	rw	
lr_undef	32	0	rw	
spsr_undef	32	0	rw	

Table 13.10: Registers at level 2, type:CPU group:Undefined

13.2.11 Abort

Registers at level:2, type:CPU group:Abort

Name	Bits	Initial-Hex	RW	Description
sp_abt	32	0	rw	
lr_abt	32	0	rw	
spsr_abt	32	0	rw	

Table 13.11: Registers at level 2, type:CPU group:Abort

13.2.12 SIMD_VFP

Registers at level:2, type:CPU group:SIMD_VFP

Name	Bits	Initial-Hex	RW	Description
d0	64	0	rw	
d1	64	0	rw	
d2	64	0	rw	
d3	64	0	rw	
d4	64	0	rw	
d5	64	0	rw	
d6	64	0	rw	
d7	64	0	rw	
d8	64	0	rw	
d9	64	0	rw	
d10	64	0	rw	
d11	64	0	rw	
d12	64	0	rw	
d13	64	0	rw	
d14	64	0	rw	
d15	64	0	rw	
d16	64	0	rw	

d17	64	0	rw	
d18	64	0	rw	
d19	64	0	rw	
d20	64	0	rw	
d21	64	0	rw	
d22	64	0	rw	
d23	64	0	rw	
d24	64	0	rw	
d25	64	0	rw	
d26	64	0	rw	
d27	64	0	rw	
d28	64	0	rw	
d29	64	0	rw	
d30	64	0	rw	
d31	64	0	rw	

Table 13.12: Registers at level 2, type:CPU group:SIMD_VFP

13.2.13 SIMD_VFP_SYS

Registers at level:2, type:CPU group:SIMD_VFP_SYS

Name	Bits	Initial-Hex	RW	Description
FPSID	32	41034080	r-	floating-point system ID
FPSCR	32	0	rw	floating-point status/control
FPExc	32	700	rw	floating-point exception
MVFR0	32	10110222	r-	Media/VFP feature 0
MVFR1	32	13211111	r-	Media/VFP feature 1
MVFR2	32	43	r-	Media/VFP feature 2

Table 13.13: Registers at level 2, type:CPU group:SIMD_VFP_SYS

13.2.14 SIMD_FP_AArch64

Registers at level:2, type:CPU group:SIMD_FP_AArch64

Name	Bits	Initial-Hex	RW	Description
v0	128	-	rw	
v1	128	-	rw	
v2	128	-	rw	
v3	128	-	rw	
v4	128	-	rw	
v5	128	-	rw	
v6	128	-	rw	
v7	128	-	rw	
v8	128	-	rw	
v9	128	-	rw	
v10	128	-	rw	
v11	128	-	rw	
v12	128	-	rw	
v13	128	-	rw	
v14	128	-	rw	
v15	128	-	rw	
v16	128	-	rw	
v17	128	-	rw	

v18	128	-	rw	
v19	128	-	rw	
v20	128	-	rw	
v21	128	-	rw	
v22	128	-	rw	
v23	128	-	rw	
v24	128	-	rw	
v25	128	-	rw	
v26	128	-	rw	
v27	128	-	rw	
v28	128	-	rw	
v29	128	-	rw	
v30	128	-	rw	
v31	128	-	rw	

Table 13.14: Registers at level 2, type:CPU group:SIMD_FP_AAarch64

13.2.15 AAarch32_32_bit_system

Registers at level:2, type:CPU group:AArch32_32_bit_system

Name	Bits	Initial-Hex	RW	Description
CNTFRQ	32	4c4b40	rw	Counter Frequency
CNTP_CTL	32	0	rw	Counter-Timer Physical Timer Control
CNTP_TVAL	32	0	rw	Counter-Timer Physical Timer TimerValue
CNTV_CTL	32	0	rw	Counter-Timer Virtual Timer Control
CNTV_TVAL	32	0	rw	Counter-Timer Virtual Timer TimerValue
CP15DMB	32	-	-w	CP15 Data Memory Barrier
CP15DSB	32	-	-w	CP15 Data Synchronization Barrier
CP15ISB	32	-	-w	CP15 Instruction Synchronization Barrier
DBGDIDR	32	3518d000	r-	Debug ID
DBGDRAR	32	0	r-	Debug ROM Address (32-bit)
DBGDSAR	32	0	r-	Debug Self Address (32-bit)
DBGDSCRint	32	0	r-	Debug Status and Control, Internal View
DBGDTRTRXint	32	0	rw	Debug Data Transfer, Transmit/Receive
DLR	32	0	rw	Debug Link
DSPSR	32	0	rw	Debug Saved Program Status
JIDR	32	0	rw	Jazelle ID
PMCCFILTR	32	0	rw	Performance Monitors Cycle Count Filter
PMCCNTR	32	0	rw	Performance Monitors Cycle Count
PMCEID0	32	fbff7f3f	r-	Performance Monitors Common Event ID 0
PMCEID1	32	1f1ae7b	r-	Performance Monitors Common Event ID 1
PMCEID2	32	0	r-	Performance Monitors Common Event ID 2
PMCEID3	32	0	r-	Performance Monitors Common Event ID 3
PMCNTENCLR	32	0	rw	Performance Monitors Count Enable Clear
PMCNTENSET	32	0	rw	Performance Monitors Count Enable Set
PMCR	32	414a3000	rw	Performance Monitors Control
PMEVCNTR0	32	0	rw	Performance Monitors Event Count 0
PMEVCNTR1	32	0	rw	Performance Monitors Event Count 1
PMEVCNTR2	32	0	rw	Performance Monitors Event Count 2
PMEVCNTR3	32	0	rw	Performance Monitors Event Count 3
PMEVCNTR4	32	0	rw	Performance Monitors Event Count 4
PMEVCNTR5	32	0	rw	Performance Monitors Event Count 5
PMEVTYPER0	32	0	rw	Performance Monitors Event Type 0
PMEVTYPER1	32	0	rw	Performance Monitors Event Type 1

PMEVTYPEPER2	32	0	rw	Performance Monitors Event Type 2
PMEVTYPEPER3	32	0	rw	Performance Monitors Event Type 3
PMEVTYPEPER4	32	0	rw	Performance Monitors Event Type 4
PMEVTYPEPER5	32	0	rw	Performance Monitors Event Type 5
PMOVSER	32	0	rw	Performance Monitors Overflow Flag Status
PMOVSSET	32	0	rw	Performance Monitors Overflow Flag Status Set
PMSELR	32	0	rw	Performance Monitors Event Counter Selection
PMSWINC	32	-	-w	Performance Monitors Software Increment
PMUSERENR	32	0	rw	Performance Monitors User Enable
PMXEVCNTR	32	0	rw	Performance Monitors Selected Event Count
PMXEVTYPER	32	0	rw	Performance Monitors Selected Event Type
TPIDRURO	32	0	rw	PL0 Read-Only Software Thread ID
TPIDRURW	32	0	rw	PL1 Software Thread ID

Table 13.15: Registers at level 2, type:CPU group:AArch32_32_bit_system

13.2.16 AArch32_32_bit_secure_system

Registers at level:2, type:CPU group:AArch32_32_bit_secure_system

Name	Bits	Initial-Hex	RW	Description
CNTP_CTL_S	32	0	rw	Counter-Timer Physical Timer Control
CNTP_TVAL_S	32	0	rw	Counter-Timer Physical Timer TimerValue
TPIDRURO_S	32	0	rw	PL0 Read-Only Software Thread ID
TPIDRURW_S	32	0	rw	PL1 Software Thread ID

Table 13.16: Registers at level 2, type:CPU group:AArch32_32_bit_secure_system

13.2.17 AArch32_32_bit_non_secure_system

Registers at level:2, type:CPU group:AArch32_32_bit_non_secure_system

Name	Bits	Initial-Hex	RW	Description
CNTP_CTL_NS	32	0	rw	Counter-Timer Physical Timer Control
CNTP_TVAL_NS	32	0	rw	Counter-Timer Physical Timer TimerValue
TPIDRURO_NS	32	0	rw	PL0 Read-Only Software Thread ID
TPIDRURW_NS	32	0	rw	PL1 Software Thread ID

Table 13.17: Registers at level 2, type:CPU group:AArch32_32_bit_non_secure_system

13.2.18 AArch32_64_bit_system

Registers at level:2, type:CPU group:AArch32_64_bit_system

Name	Bits	Initial-Hex	RW	Description
CNTPCT	64	0	r-	Counter-Timer Physical Count
CNTP_CVAL	64	0	rw	Counter-Timer Physical Timer CompareValue
CNTVCT	64	0	r-	Counter-Timer Virtual Count
CNTV_CVAL	64	0	rw	Counter-Timer Virtual Timer CompareValue
DBGDRAR64	64	0	r-	Debug ROM Address (64-bit)
DBGDSAR64	64	0	r-	Debug Self Address (64-bit)
PMCCNTR64	64	0	rw	Performance Monitors Cycle Count (64-bit)

Table 13.18: Registers at level 2, type:CPU group:AArch32_64_bit_system

13.2.19 AArch32_64_bit_secure_system

Registers at level:2, type:CPU group:AArch32_64_bit_secure_system

Name	Bits	Initial-Hex	RW	Description
CNTP_CVAL_S	64	0	rw	Counter-Timer Physical Timer CompareValue

Table 13.19: Registers at level 2, type:CPU group:AArch32_64_bit_secure_system

13.2.20 AArch32_64_bit_non_secure_system

Registers at level:2, type:CPU group:AArch32_64_bit_non_secure_system

Name	Bits	Initial-Hex	RW	Description
CNTP_CVAL_NS	64	0	rw	Counter-Timer Physical Timer CompareValue

Table 13.20: Registers at level 2, type:CPU group:AArch32_64_bit_non_secure_system

13.2.21 AArch64_system

Registers at level:2, type:CPU group:AArch64_system

Name	Bits	Initial-Hex	RW	Description
ACTLR_EL1	64	0	rw	Auxiliary Control (EL1)
ACTLR_EL2	64	0	rw	Auxiliary Control (EL2)
ACTLR_EL3	64	0	rw	Auxiliary Control (EL3)
AFSR0_EL1	32	0	rw	Auxiliary Fault Status 0 (EL1)
AFSR0_EL2	32	0	rw	Auxiliary Fault Status 0 (EL2)
AFSR0_EL3	32	0	rw	Auxiliary Fault Status 0 (EL3)
AFSR0_EL12	32	0	rw	Auxiliary Fault Status 0 (EL1) Alias
AFSR1_EL1	32	0	rw	Auxiliary Fault Status 1 (EL1)
AFSR1_EL2	32	0	rw	Auxiliary Fault Status 1 (EL2)
AFSR1_EL3	32	0	rw	Auxiliary Fault Status 1 (EL3)
AFSR1_EL12	32	0	rw	Auxiliary Fault Status 1 (EL1) Alias
AIDR_EL1	32	0	r-	Auxiliary ID
AMAIR_EL1	64	0	rw	Auxiliary Memory Attribute Indirection (EL1)
AMAIR_EL2	64	0	rw	Auxiliary Memory Attribute Indirection (EL2)
AMAIR_EL3	64	0	rw	Auxiliary Memory Attribute Indirection (EL3)
AMAIR_EL12	64	0	rw	Auxiliary Memory Attribute Indirection (EL1) Alias
ATCR_EL1	32	0	rw	Auxiliary Translation Control (EL1)
ATCR_EL2	32	0	rw	Auxiliary Translation Control (EL2)
ATCR_EL3	32	0	rw	Auxiliary Translation Control (EL3)
ATCR_EL12	64	0	rw	Auxiliary Translation Control (EL1) Alias
AVTCR_EL2	32	0	rw	Auxiliary Virtualized Translation Control (EL2)
CCSIDR_EL1	32	701fe01a	r-	Current Cache Size ID
CLIDR_EL1	32	c3000123	r-	Cache Level ID
CLUSTERACPSID_EL1	32	0	rw	Cluster ACP Scheme ID
CLUSTERACTLR_EL1	32	0	rw	Cluster Auxiliary Control
CLUSTERBUSQOS_EL1	32	eeeeeeee	rw	Cluster Bus QoS Control
CLUSTERCFR_EL1	32	3d093	rw	Cluster Configuration
CLUSTERECTLR_EL1	32	500	rw	Cluster Extended Control

CLUSTERIDR_EL1	32	2	rw	Cluster Main Revision ID
CLUSTERL3HIT_EL1	32	0	rw	Cluster L3 Hit Counter
CLUSTERL3MISS_EL1	32	0	rw	Cluster L3 Miss Counter
CLUSTERPARTCR_EL1	32	0	rw	Cluster Partition Control
CLUSTERPMCCTR_EL1	64	0	rw	Cluster Performance Monitors Cycle Count
CLUSTERPMCEID0_EL1	32	66020000	r-	Cluster Common Event ID 0
CLUSTERPMCEID1_EL1	32	1e00	r-	Cluster Common Event ID 1
CLUSTERPMCLAIMCLR_EL1	32	0	rw	Cluster Performance Monitor Claim Tag Clear
CLUSTERPMCLAIMSET_EL1	32	0	rw	Cluster Performance Monitor Claim Tag Set
CLUSTERPMCNTENCLR_EL1	32	0	rw	Cluster Count Enable Clear
CLUSTERPMCNTENSET_EL1	32	0	rw	Cluster Count Enable Set
CLUSTERPMCR_EL1	32	41413040	rw	Cluster Performance Monitors Control
CLUSTERPMINTENCLR_EL1	32	0	rw	Cluster Interrupt Enable Clear
CLUSTERPMINTENSET_EL1	32	0	rw	Cluster Interrupt Enable Set
CLUSTERPMMDCR_EL3	32	0	rw	Cluster Monitor Debug Configuration
CLUSTERPMOVSCCLR_EL1	32	0	rw	Cluster Overflow Flag Status Clear
CLUSTERPMOVSSSET_EL1	32	0	rw	Cluster Overflow Flag Status Set
CLUSTERPMSELR_EL1	32	0	rw	Cluster Event Counter Selection
CLUSTERPMXEVCNTR_EL1	32	0	rw	Cluster Selected Event Count
CLUSTERPMXEVTYPYPER_EL1	32	0	rw	Cluster Selected Event Type
CLUSTERPWRCTLR_EL1	32	0	rw	Cluster Power Control
CLUSTERPWRDN_EL1	32	0	rw	Cluster Powerdown
CLUSTERPWRSTAT_EL1	32	0	rw	Cluster Power Status
CLUSTERREVIDR_EL1	32	0	rw	Cluster Revision ID
CLUSTERSTASHSID_EL1	32	0	rw	Cluster Stash Scheme ID
CLUSTERTHREADSIDOVR_EL1	32	0	rw	Cluster Thread Scheme ID Override
CLUSTERTHREADSID_EL1	32	0	rw	Cluster Thread Scheme ID
CNTFRQ_EL0	32	4c4b40	rw	Counter-Timer Frequency
CNTHCTL_EL2	32	3	rw	Counter-Timer Hypervisor Control
CNTHP_CTL_EL2	32	0	rw	Counter-Timer Physical Timer Control (EL2)
CNTHP_CVAL_EL2	64	0	rw	Counter-Timer Physical Timer CompareValue (EL2)
CNTHP_TVAL_EL2	32	0	rw	Counter-Timer Physical Timer TimerValue (EL2)
CNTHV_CTL_EL2	32	0	rw	Counter-Timer Virtual Timer Control (EL2)
CNTHV_CVAL_EL2	64	0	rw	Counter-Timer Virtual Timer CompareValue (EL2)
CNTHV_TVAL_EL2	32	0	rw	Counter-Timer Virtual Timer TimerValue (EL2)
CNTKCTL_EL1	32	0	rw	Counter-Timer Kernel Control
CNTKCTL_EL12	32	0	rw	Counter-Timer Kernel Control Alias
CNTPCT_EL0	64	0	r-	Counter-Timer Physical Count
CNTPS_CTL_EL1	32	0	rw	Counter-Timer Secure Physical Timer Control
CNTPS_CVAL_EL1	64	0	rw	Counter-Timer Secure Physical Timer CompareValue
CNTPS_TVAL_EL1	32	0	rw	Counter-Timer Secure Physical Timer TimerValue
CNTP_CTL_EL0	32	0	rw	Counter-Timer Physical Timer Control
CNTP_CTL_EL02	32	0	rw	Counter-Timer Physical Timer Control Alias
CNTP_CVAL_EL0	64	0	rw	Counter-Timer Physical Timer CompareValue

CNTP_CVAL_EL02	64	0	rw	Counter-Timer Physical Timer CompareValue Alias
CNTP_TVAL_EL0	32	0	rw	Counter-Timer Physical Timer TimerValue
CNTP_TVAL_EL02	32	0	rw	Counter-Timer Physical Timer TimerValue Alias
CNTVCT_EL0	64	0	r-	Counter-Timer Virtual Count
CNTVOFF_EL2	64	0	rw	Counter-Timer Virtual Offset
CNTV_CTL_EL0	32	0	rw	Counter-Timer Virtual Timer Control
CNTV_CTL_EL02	32	0	rw	Counter-Timer Virtual Timer Control Alias
CNTV_CVAL_EL0	64	0	rw	Counter-Timer Virtual Timer CompareValue
CNTV_CVAL_EL02	64	0	rw	Counter-Timer Virtual Timer CompareValue Alias
CNTV_TVAL_EL0	32	0	rw	Counter-Timer Virtual Timer TimerValue
CNTV_TVAL_EL02	32	0	rw	Counter-Timer Virtual Timer TimerValue Alias
CONTEXTIDR_EL1	32	0	rw	Context ID (EL1)
CONTEXTIDR_EL2	32	0	rw	Context ID (EL2)
CONTEXTIDR_EL12	32	0	rw	Context ID (EL1) Alias
CPACR_EL1	32	0	rw	Architectural Feature Access Control
CPACR_EL12	32	0	rw	Architectural Feature Access Control Alias
CPTR_EL2	32	33ff	rw	Architectural Feature Trap (EL2)
CPTR_EL3	32	0	rw	Architectural Feature Trap (EL3)
CPUACTLR2_EL1	64	0	rw	CPU Auxiliary Control 2
CPUACTLR3_EL1	64	0	rw	CPU Auxiliary Control 3
CPUACTLR_EL1	64	0	rw	CPU Auxiliary Control
CPUAMCFGR_EL0	32	3f04	r-	Activity Monitors Configuration
CPUAMCNTENCLR_EL0	32	0	rw	Activity Monitor Enable Clear
CPUAMCNTENSET_EL0	32	0	rw	Activity Monitor Enable Set
CPUAMEVCNTR0_EL0	64	0	rw	Activity Monitor Event Counter 0
CPUAMEVCNTR1_EL0	64	0	rw	Activity Monitor Event Counter 1
CPUAMEVCNTR2_EL0	64	0	rw	Activity Monitor Event Counter 2
CPUAMEVCNTR3_EL0	64	0	rw	Activity Monitor Event Counter 3
CPUAMEVCNTR4_EL0	64	0	rw	Activity Monitor Event Counter 4
CPUAMEVTYPER0_EL0	32	11	rw	Activity Monitor Event Type 0
CPUAMEVTYPER1_EL0	32	ef	rw	Activity Monitor Event Type 1
CPUAMEVTYPER2_EL0	32	8	rw	Activity Monitor Event Type 2
CPUAMEVTYPER3_EL0	32	0	rw	Activity Monitor Event Type 3
CPUAMEVTYPER4_EL0	32	0	rw	Activity Monitor Event Type 4
CPUAMUSERENR_EL0	32	0	rw	Activity Monitor EL0 Enable Access
CPUCFR_EL1	32	0	r-	CPU Configuration
CPUECTLR_EL1	64	9 61563000	rw	CPU Extended Control
CPUPCR_EL3	64	0	rw	CPU Private Control
CPUPMR_EL3	64	0	rw	CPU Private Mask
CPUPOR_EL3	64	0	rw	CPU Private Operation
CPUPSELR_EL3	32	0	rw	CPU Private Selection
CPUPWRCTLR_EL1	32	0	rw	CPU Power Control
CSSELR_EL1	32	0	rw	Current Size Selection
CTR_EL0	32	8444c004	r-	Cache Type
CurrentEL	32	c	r-	Current Exception Level
DAIF	32	3c0	rw	Interrupt Mask Bits
DBGAUTHSTATUS_EL1	32	aa	r-	Debug Authentication Status
DBGBCR0_EL1	32	0	rw	Debug Breakpoint Control 0
DBGBCR1_EL1	32	0	rw	Debug Breakpoint Control 1
DBGBCR2_EL1	32	0	rw	Debug Breakpoint Control 2
DBGBCR3_EL1	32	0	rw	Debug Breakpoint Control 3

DBGBCR4_EL1	32	0	rw	Debug Breakpoint Control 4
DBGBCR5_EL1	32	0	rw	Debug Breakpoint Control 5
DBGBVR0_EL1	64	0	rw	Debug Breakpoint Value 0
DBGBVR1_EL1	64	0	rw	Debug Breakpoint Value 1
DBGBVR2_EL1	64	0	rw	Debug Breakpoint Value 2
DBGBVR3_EL1	64	0	rw	Debug Breakpoint Value 3
DBGBVR4_EL1	64	0	rw	Debug Breakpoint Value 4
DBGBVR5_EL1	64	0	rw	Debug Breakpoint Value 5
DBGCLAIMCLR_EL1	32	0	rw	Debug Claim Tag Clear
DBGCLAIMSET_EL1	32	0	rw	Debug Claim Tag Set
DBGDTRTRX_EL0	32	0	rw	Debug Data Transfer, Transmit/Receive
DBGDTR_EL0	64	0	rw	Debug Data Transfer
DBGPRCR_EL1	32	0	rw	Debug Power Control
DBGWCR0_EL1	32	0	rw	Debug Watchpoint Control 0
DBGWCR1_EL1	32	0	rw	Debug Watchpoint Control 1
DBGWCR2_EL1	32	0	rw	Debug Watchpoint Control 2
DBGWCR3_EL1	32	0	rw	Debug Watchpoint Control 3
DBGWVR0_EL1	64	0	rw	Debug Watchpoint Value 0
DBGWVR1_EL1	64	0	rw	Debug Watchpoint Value 1
DBGWVR2_EL1	64	0	rw	Debug Watchpoint Value 2
DBGWVR3_EL1	64	0	rw	Debug Watchpoint Value 3
DCZID_EL0	32	4	r-	Data Cache Zero ID
DISR_EL1	32	0	rw	Deferred Interrupt Status
DLR_EL0	64	0	rw	Debug Link
DSPSR_EL0	32	0	rw	Debug Saved Program Status
ELR_EL1	64	0	rw	Exception Link (EL1)
ELR_EL2	64	0	rw	Exception Link (EL2)
ELR_EL3	64	0	rw	Exception Link (EL3)
ELR_EL12	32	0	rw	Exception Link (EL1) Alias
ERRIDR_EL1	32	0	r-	Error Record ID
ERRSELR_EL1	32	0	rw	Error Record Select
ERXADDR_EL1	64	0	rw	Selected Error Record Address
ERXCTLR_EL1	64	0	rw	Selected Error Record Control
ERXFR_EL1	64	0	r-	Selected Error Record Feature
ERXMISC0_EL1	64	0	rw	Selected Error Record Miscellaneous 0
ERXMISC1_EL1	64	0	rw	Selected Error Record Miscellaneous 1
ERXPFGCDNR_EL1	32	0	rw	Selected Error Pseudo Fault Generation Count Down
ERXPFGCTLR_EL1	32	0	rw	Selected Error Pseudo Fault Generation Control
ERXPFGFR_EL1	32	0	r-	Selected Error Pseudo Fault Generation Feature
ERXSTATUS_EL1	32	0	rw	Selected Error Record Status
ESR_EL1	32	0	rw	Exception Syndrome (EL1)
ESR_EL2	32	0	rw	Exception Syndrome (EL2)
ESR_EL3	32	0	rw	Exception Syndrome (EL3)
ESR_EL12	32	0	rw	Exception Syndrome (EL1) Alias
FAR_EL1	64	0	rw	Fault Address (EL1)
FAR_EL2	64	0	rw	Fault Address (EL2)
FAR_EL3	64	0	rw	Fault Address (EL3)
FAR_EL12	64	0	rw	Fault Address (EL1) Alias
FPCR	32	0	rw	Floating Point Control
FPSR	32	0	rw	Floating Point Status
HACR_EL2	32	0	rw	Hypervisor Auxiliary Control
HCR_EL2	64	80000002	rw	Hypervisor Configuration
HPFAR_EL2	64	0	rw	Hypervisor IPA Fault Address

HSTR_EL2	32	0	rw	Hypervisor System Trap
ICC_AP0R0_EL1	32	0	rw	Interrupt Controller Active Priorities Group 0, 0
ICC_AP1R0_EL1	32	0	rw	Interrupt Controller Active Priorities Group 1, 0
ICC_ASGI1R_EL1	64	-	-w	Interrupt Controller Alias SGI Group 1
ICC_BPR0_EL1	32	2	rw	Interrupt Controller Binary Point 0
ICC_BPR1_EL1	32	2	rw	Interrupt Controller Binary Point 1
ICC_CTLR_EL1	32	400	rw	Interrupt Controller Control (EL1)
ICC_CTLR_EL3	32	400	rw	Interrupt Controller Control (EL3)
ICC_DIR_EL1	32	-	-w	Interrupt Controller Deactivate Interrupt
ICC_EOIR0_EL1	32	-	-w	Interrupt Controller End of Interrupt 0
ICC_EOIR1_EL1	32	-	-w	Interrupt Controller End of Interrupt 1
ICC_HPPIR0_EL1	32	3ff	r-	Interrupt Controller Highest Priority Pending Interrupt 0
ICC_HPPIR1_EL1	32	3ff	r-	Interrupt Controller Highest Priority Pending Interrupt 1
ICC_IAR0_EL1	32	3ff	r-	Interrupt Controller Interrupt Acknowledge 0
ICC_IAR1_EL1	32	3ff	r-	Interrupt Controller Interrupt Acknowledge 1
ICC_IGRPEN0_EL1	32	0	rw	Interrupt Controller Interrupt Group 0 Enable
ICC_IGRPEN1_EL1	32	0	rw	Interrupt Controller Interrupt Group 1 Enable
ICC_IGRPEN1_EL3	32	0	rw	Interrupt Controller Interrupt Group 1 Enable (EL3)
ICC_PMR_EL1	32	0	rw	Interrupt Controller Priority Mask
ICC_RPR_EL1	32	ff	r-	Interrupt Controller Running Priority
ICC_SGI0R_EL1	64	-	-w	Interrupt Controller SGI Group 0
ICC_SGI1R_EL1	64	-	-w	Interrupt Controller SGI Group 1
ICC_SRE_EL1	32	1	rw	Interrupt Controller System Register Enable (EL1)
ICC_SRE_EL2	32	9	rw	Interrupt Controller System Register Enable (EL2)
ICC_SRE_EL3	32	9	rw	Interrupt Controller System Register Enable (EL3)
ICH_AP0R0_EL2	32	0	rw	Interrupt Controller Hyp Active Priorities Group 0 (Word 0)
ICH_AP1R0_EL2	32	0	rw	Interrupt Controller Hyp Active Priorities Group 1 (Word 0)
ICH_EISR_EL2	32	0	r-	Interrupt Controller End of Interrupt Status
ICH_ELRSR_EL2	32	f	r-	Interrupt Controller Empty List Register Status
ICH_HCR_EL2	32	0	rw	Interrupt Controller Hypervisor Control
ICH_LR0_EL2	64	0	rw	Interrupt Controller List 0
ICH_LR1_EL2	64	0	rw	Interrupt Controller List 1
ICH_LR2_EL2	64	0	rw	Interrupt Controller List 2
ICH_LR3_EL2	64	0	rw	Interrupt Controller List 3
ICH_MISR_EL2	32	0	r-	Interrupt Controller Maintenance Interrupt State
ICH_VMCR_EL2	32	4c0000	rw	Interrupt Controller Virtual Machine Control
ICH_VTR_EL2	32	90100003	r-	Interrupt Controller VGIC Type
ID_AA64AFR0_EL1	64	0	r-	AArch64 Auxiliary Feature 0
ID_AA64AFR1_EL1	64	0	r-	AArch64 Auxiliary Feature 1

ID_AA64DFR0_EL1	64	10305408	r-	AArch64 Debug Feature 0
ID_AA64DFR1_EL1	64	0	r-	AArch64 Debug Feature 1
ID_AA64ISAR0_EL1	64	1000 10211120	r-	AArch64 Instruction Set Attribute 0
ID_AA64ISAR1_EL1	64	100001	r-	AArch64 Instruction Set Attribute 1
ID_AA64MMFR0_EL1	64	101122	r-	AArch64 Memory Model Feature 0
ID_AA64MMFR1_EL1	64	10212122	r-	AArch64 Memory Model Feature 1
ID_AA64MMFR2_EL1	64	1011	r-	AArch64 Memory Model Feature 2
ID_AA64PFR0_EL1	64	11000000 11111112	r-	AArch64 Processor Feature 0
ID_AA64PFR1_EL1	64	10	r-	AArch64 Processor Feature 1
ID_AFR0_EL1	32	0	r-	Auxiliary Feature 0
ID_DFR0_EL1	32	4010088	r-	Debug Feature 0
ID_ISAR0_EL1	32	2101110	r-	Instruction Set Attribute 0
ID_ISAR1_EL1	32	13112111	r-	Instruction Set Attribute 1
ID_ISAR2_EL1	32	21232042	r-	Instruction Set Attribute 2
ID_ISAR3_EL1	32	1112131	r-	Instruction Set Attribute 3
ID_ISAR4_EL1	32	10142	r-	Instruction Set Attribute 4
ID_ISAR5_EL1	32	1011121	r-	Instruction Set Attribute 5
ID_ISAR6_EL1	32	10	r-	Instruction Set Attribute 6
ID_MMFR0_EL1	32	10201105	r-	Memory Model Feature 0
ID_MMFR1_EL1	32	40000000	r-	Memory Model Feature 1
ID_MMFR2_EL1	32	1260000	r-	Memory Model Feature 2
ID_MMFR3_EL1	32	2122211	r-	Memory Model Feature 3
ID_MMFR4_EL1	32	21110	r-	Memory Model Feature 4
ID_PFR0_EL1	32	10010131	r-	Processor Feature 0
ID_PFR1_EL1	32	10010000	r-	Processor Feature 1
ID_PFR2_EL1	32	11	r-	Processor Feature 2
ISR_EL1	32	0	r-	Interrupt Status
LORC_EL1	32	0	rw	LORegion Control
LOREA_EL1	64	0	rw	LORegion End Address
LORID_EL1	32	40004	r-	LORegion ID
LORN_EL1	32	0	rw	LORegion Number
LORSA_EL1	64	0	rw	LORegion Start Address
MAIR_EL1	64	44e048e0 00098aa4	rw	Memory Attribute Indirection (EL1)
MAIR_EL2	64	0	rw	Memory Attribute Indirection (EL2)
MAIR_EL3	64	44e048e0 00098aa4	rw	Memory Attribute Indirection (EL3)
MAIR_EL12	64	0	rw	Memory Attribute Indirection (EL1) Alias
MDCCINT_EL1	32	0	rw	Monitor DCC Interrupt Enable
MDCCSR_EL0	32	0	r-	Monitor DCC Status
MDCR_EL2	32	6	rw	Monitor Debug Configuration (EL2)
MDCR_EL3	32	0	rw	Monitor Debug Configuration (EL3)
MDRAR_EL1	64	0	r-	Monitor Debug ROM Address
MDSCR_EL1	32	0	rw	Monitor Debug System Control
MIDR_EL1	32	414fd0b1	r-	Main ID
MPIDR_EL1	64	81000000	r-	Multiprocessor Affinity
MVFR0_EL1	32	10110222	r-	Media and VFP Feature 0
MVFR1_EL1	32	13211111	r-	Media and VFP Feature 1
MVFR2_EL1	32	43	r-	Media and VFP Feature 2
NZCV	32	0	rw	Condition Flags
OSDLR_EL1	32	0	rw	OS Double Lock
OSDTRRX_EL1	32	0	rw	OS Lock Data Transfer, Receive
OSDTRTX_EL1	32	0	rw	OS Lock Data Transfer, Transmit
OSECCR_EL1	32	0	rw	OS Lock Exception Catch Control

OSLAR_EL1	32	-	-w	OS Lock Access
OSLSR_EL1	32	a	r-	OS Lock Status
PAN	32	0	rw	Privileged Access Never
PAR_EL1	64	0	rw	Physical Address
PMCCFILTR_EL0	32	0	rw	Performance Monitors Cycle Count Filter
PMCCNTR_EL0	64	0	rw	Performance Monitors Cycle Count
PMCEID0_EL0	64	fbff7f3f	r-	Performance Monitors Common Event ID 0
PMCEID1_EL0	64	1f1ae7b	r-	Performance Monitors Common Event ID 1
PMCNTENCLR_EL0	32	0	rw	Performance Monitors Count Enable Clear
PMCNTENSET_EL0	32	0	rw	Performance Monitors Count Enable Set
PMCR_EL0	32	414a3000	rw	Performance Monitors Control
PMEVCNTR0_EL0	32	0	rw	Performance Monitors Event Count 0
PMEVCNTR1_EL0	32	0	rw	Performance Monitors Event Count 1
PMEVCNTR2_EL0	32	0	rw	Performance Monitors Event Count 2
PMEVCNTR3_EL0	32	0	rw	Performance Monitors Event Count 3
PMEVCNTR4_EL0	32	0	rw	Performance Monitors Event Count 4
PMEVCNTR5_EL0	32	0	rw	Performance Monitors Event Count 5
PMEVTYPE0_EL0	32	0	rw	Performance Monitors Event Type 0
PMEVTYPE1_EL0	32	0	rw	Performance Monitors Event Type 1
PMEVTYPE2_EL0	32	0	rw	Performance Monitors Event Type 2
PMEVTYPE3_EL0	32	0	rw	Performance Monitors Event Type 3
PMEVTYPE4_EL0	32	0	rw	Performance Monitors Event Type 4
PMEVTYPE5_EL0	32	0	rw	Performance Monitors Event Type 5
PMINTENCLR_EL1	32	0	rw	Performance Monitors Interrupt Enable Clear
PMINTENSET_EL1	32	0	rw	Performance Monitors Interrupt Enable Set
PMOVSCLR_EL0	32	0	rw	Performance Monitors Overflow Flag Status Clear
PMOVSSET_EL0	32	0	rw	Performance Monitors Overflow Flag Status Set
PMSELR_EL0	32	0	rw	Performance Monitors Event Counter Selection
PMSWINC_EL0	32	-	-w	Performance Monitors Software Increment
PMUSERENR_EL0	32	0	rw	Performance Monitors User Enable
PMXEVCNTR_EL0	32	0	rw	Performance Monitors Selected Event Count
PMXEVTYPER_EL0	32	0	rw	Performance Monitors Selected Event Type
REVIDR_EL1	32	0	r-	Revision ID
RVBAR_EL3	64	0	r-	Reset Vector Base Address (EL3)
SCR_EL3	32	430	rw	Secure Configuration
SCTLR_EL1	64	30d50838	rw	System Control (EL1)
SCTLR_EL2	64	30c50838	rw	System Control (EL2)
SCTLR_EL3	64	30c50838	rw	System Control (EL3)
SCTLR_EL12	64	0	rw	System Control (EL1) Alias
SPSR_EL1	32	0	rw	Saved Program Status (EL1)
SPSR_EL2	32	0	rw	Saved Program Status (EL2)
SPSR_EL3	32	0	rw	Saved Program Status (EL3)
SPSR_EL12	32	0	rw	Saved Program Status (EL1) Alias
SPSR_abt	32	0	rw	Saved Program Status (Abort Mode)
SPSR_fiq	32	0	rw	Saved Program Status (FIQ Mode)
SPSR_irq	32	0	rw	Saved Program Status (IRQ Mode)
SPSR_und	32	0	rw	Saved Program Status (Undefined Mode)
SPSel	32	1	rw	Stack Pointer Select
SP_EL0	64	0	rw	Stack Pointer (EL0)
SP_EL1	64	0	rw	Stack Pointer (EL1)
SP_EL2	64	0	rw	Stack Pointer (EL2)
SP_EL3	64	0	rw	Stack Pointer (EL3)

TCR_EL1	64	0	rw	Translation Control (EL1)
TCR_EL2	64	80000000	rw	Translation Control (EL2)
TCR_EL3	32	80800000	rw	Translation Control (EL3)
TCR_EL12	64	0	rw	Translation Control (EL1) Alias
TPIDRRO_EL0	64	0	rw	Thread Pointer/ID, Read-Only (EL0)
TPIDR_EL0	64	0	rw	Thread Pointer/ID (EL0)
TPIDR_EL1	64	0	rw	Thread Pointer/ID (EL1)
TPIDR_EL2	64	0	rw	Thread Pointer/ID (EL2)
TPIDR_EL3	64	0	rw	Thread Pointer/ID (EL3)
TTBR0_EL1	64	0	rw	Translation Table Base 0 (EL1)
TTBR0_EL2	64	0	rw	Translation Table Base 0 (EL2)
TTBR0_EL3	64	0	rw	Translation Table Base 0 (EL3)
TTBR0_EL12	64	0	rw	Translation Table Base 0 (EL1) Alias
TTBR1_EL1	64	0	rw	Translation Table Base 1 (EL1)
TTBR1_EL2	64	0	rw	Translation Table Base 1 (EL2)
TTBR1_EL12	64	0	rw	Translation Table Base 1 (EL1) Alias
UAO	32	0	rw	Unprivileged Access Override
VBAR_EL1	64	0	rw	Vector Base Address (EL1)
VBAR_EL2	64	0	rw	Vector Base Address (EL2)
VBAR_EL3	64	0	rw	Vector Base Address (EL3)
VBAR_EL12	64	0	rw	Vector Base Address (EL1) Alias
VDISR_EL2	32	0	rw	Virtual Deferred Interrupt Status
VMPIDR_EL2	64	81000000	rw	Virtualization Multiprocessor ID
VPIDR_EL2	32	414fd0b1	rw	Virtualization Processor ID
VSESR_EL2	32	0	rw	Virtual SError Exception Syndrome
VTCR_EL2	32	80000000	rw	Virtualization Translation Control
VTTBR_EL2	64	0	rw	Virtualization Translation Table Base

Table 13.21: Registers at level 2, type:CPU group:AArch64_system

13.2.22 AArch64_secure_system

Registers at level:2, type:CPU group:AArch64_secure_system

Name	Bits	Initial-Hex	RW	Description
ICC_APIR0_EL1.S	32	0	rw	Interrupt Controller Active Priorities Group 1, 0
ICC_ASGI1R_EL1.S	64	-	-w	Interrupt Controller Alias SGI Group 1
ICC_BPRI_EL1.S	32	2	rw	Interrupt Controller Binary Point 1
ICC_CTLR_EL1.S	32	400	rw	Interrupt Controller Control (EL1)
ICC_DIR_EL1.S	32	-	-w	Interrupt Controller Deactivate Interrupt
ICC_EOIR0_EL1.S	32	-	-w	Interrupt Controller End of Interrupt 0
ICC_EOIR1_EL1.S	32	-	-w	Interrupt Controller End of Interrupt 1
ICC_HPPIR0_EL1.S	32	3ff	r-	Interrupt Controller Highest Priority Pending Interrupt 0
ICC_HPPIR1_EL1.S	32	3ff	r-	Interrupt Controller Highest Priority Pending Interrupt 1
ICC_IAR0_EL1.S	32	3ff	r-	Interrupt Controller Interrupt Acknowledge 0
ICC_IAR1_EL1.S	32	3ff	r-	Interrupt Controller Interrupt Acknowledge 1
ICC_IGRPEN0_EL1.S	32	0	rw	Interrupt Controller Interrupt Group 0 Enable
ICC_IGRPEN1_EL1.S	32	0	rw	Interrupt Controller Interrupt Group 1 Enable
ICC_IGRPEN1_EL3.S	32	0	rw	Interrupt Controller Interrupt Group 1 Enable (EL3)
ICC_PMR_EL1.S	32	0	rw	Interrupt Controller Priority Mask
ICC_RPR_EL1.S	32	ff	r-	Interrupt Controller Running Priority
ICC_SGI0R_EL1.S	64	-	-w	Interrupt Controller SGI Group 0
ICC_SGI1R_EL1.S	64	-	-w	Interrupt Controller SGI Group 1
ICC_SRE_EL1.S	32	1	rw	Interrupt Controller System Register Enable (EL1)

Table 13.22: Registers at level 2, type:CPU group:AArch64_secure_system

13.2.23 AArch64_non_secure_system

Registers at level:2, type:CPU group:AArch64_non_secure_system

Name	Bits	Initial-Hex	RW	Description
ICC_AP1R0_EL1_NS	32	0	rw	Interrupt Controller Active Priorities Group 1, 0
ICC_ASGI1R_EL1_NS	64	-	-w	Interrupt Controller Alias SGI Group 1
ICC_BPR1_EL1_NS	32	3	rw	Interrupt Controller Binary Point 1
ICC_CTLR_EL1_NS	32	400	rw	Interrupt Controller Control (EL1)
ICC_DIR_EL1_NS	32	-	-w	Interrupt Controller Deactivate Interrupt
ICC_EOIR0_EL1_NS	32	-	-w	Interrupt Controller End of Interrupt 0
ICC_EOIR1_EL1_NS	32	-	-w	Interrupt Controller End of Interrupt 1
ICC_HPPIR0_EL1_NS	32	3ff	r-	Interrupt Controller Highest Priority Pending Interrupt 0
ICC_HPPIR1_EL1_NS	32	3ff	r-	Interrupt Controller Highest Priority Pending Interrupt 1
ICC_IAR0_EL1_NS	32	3ff	r-	Interrupt Controller Interrupt Acknowledge 0
ICC_IAR1_EL1_NS	32	3ff	r-	Interrupt Controller Interrupt Acknowledge 1
ICC_IGRPEN0_EL1_NS	32	0	rw	Interrupt Controller Interrupt Group 0 Enable
ICC_IGRPEN1_EL1_NS	32	0	rw	Interrupt Controller Interrupt Group 1 Enable
ICC_IGRPEN1_EL3_NS	32	0	rw	Interrupt Controller Interrupt Group 1 Enable (EL3)
ICC_PMR_EL1_NS	32	0	rw	Interrupt Controller Priority Mask
ICC_RPR_EL1_NS	32	ff	r-	Interrupt Controller Running Priority
ICC_SGI0R_EL1_NS	64	-	-w	Interrupt Controller SGI Group 0
ICC_SGI1R_EL1_NS	64	-	-w	Interrupt Controller SGI Group 1
ICC_SRE_EL1_NS	32	1	rw	Interrupt Controller System Register Enable (EL1)

Table 13.23: Registers at level 2, type:CPU group:AArch64_non_secure_system

13.2.24 AArch64_SYS instruction registers

Registers at level:2, type:CPU group:AArch64_SYS_instruction_registers

Name	Bits	Initial-Hex	RW	Description
ATS1E0R	64	-	-w	
ATS1E0W	64	-	-w	
ATS1E1R	64	-	-w	
ATS1E1RP	64	-	-w	
ATS1E1W	64	-	-w	
ATS1E1WP	64	-	-w	
ATS1E2R	64	-	-w	
ATS1E2W	64	-	-w	
ATS1E3R	64	-	-w	
ATS1E3W	64	-	-w	
ATS12E0R	64	-	-w	
ATS12E0W	64	-	-w	
ATS12E1R	64	-	-w	
ATS12E1W	64	-	-w	
DCCISW	32	-	-w	
DCCIVAC	64	-	-w	
DCCSW	32	-	-w	
DCCVAC	64	-	-w	
DCCVAP	64	-	-w	
DCCVAU	64	-	-w	
DCISW	32	-	-w	
DCIVAC	64	-	-w	
DCZVA	32	-	-w	
ICIALLU	32	-	-w	

ICIALUIS	32	-	-w	
ICIVAU	64	-	-w	
TLBIALLE1	64	-	-w	
TLBIALLE1IS	64	-	-w	
TLBIALLE2	64	-	-w	
TLBIALLE2IS	64	-	-w	
TLBIALLE3	64	-	-w	
TLBIALLE3IS	64	-	-w	
TLBIASIDE1	64	-	-w	
TLBIASIDE1IS	64	-	-w	
TLBIIPAS2E1	64	-	-w	
TLBIIPAS2E1IS	64	-	-w	
TLBIIPAS2LE1	64	-	-w	
TLBIIPAS2LE1IS	64	-	-w	
TLBIVAAE1	64	-	-w	
TLBIVAAE1IS	64	-	-w	
TLBIVAALE1	64	-	-w	
TLBIVAALE1IS	64	-	-w	
TLBIVAE1	64	-	-w	
TLBIVAE1IS	64	-	-w	
TLBIVAE2	64	-	-w	
TLBIVAE2IS	64	-	-w	
TLBIVAE3	64	-	-w	
TLBIVAE3IS	64	-	-w	
TLBIVALE1	64	-	-w	
TLBIVALE1IS	64	-	-w	
TLBIVALE2	64	-	-w	
TLBIVALE2IS	64	-	-w	
TLBIVALE3	64	-	-w	
TLBIVALE3IS	64	-	-w	
TLBIVMALE1	64	-	-w	
TLBIVMALE1IS	64	-	-w	
TLBIVMALLS12E1	64	-	-w	
TLBIVMALLS12E1IS	64	-	-w	

Table 13.24: Registers at level 2, type:CPU group:AArch64_SYS_instruction_registers

13.2.25 Integration_support

Registers at level:2, type:CPU group:Integration_support

Name	Bits	Initial-Hex	RW	Description
transactPL	32	3	r-	privilege level of current memory transaction
transactAT	32	0	r-	current memory transaction type: PA=1, VA=0
artifactPAR	64	0	r-	result of address translation for artifact write to ATS1CPR etc
PTWBankSelect	8	0	rw	select PTW bank (0 is stage 1, 1 is stage 2, 2-5 are stage 2 walks initiated by stage 1 level 0-3 entry lookups, respectively)
PTWBankValid	8	0	r-	bitmask of valid banks (0x01 is stage 1, 0x02 is stage 2, 0x04-0x20 are stage 2 walks initiated by stage 1 level 0-3 entry lookups, respectively)
PTWAddressValid	8	0	r-	bitmask of valid bits for each of PTWAddressL0...PTWAddressL3, PTWBase, PTWInput and PTWOutput in current bank

PTWAddressNS	8	0	r-	bitmask of Non-Secure bits for each of PTWAddressL0...PTWAddressL3, PTWBase and PTWOutput in current bank (PTWInput bit is always 0)
PTWValueValid	8	0	r-	bitmask of valid bits for each of PTWValueL0...PTWValueL3 in current bank
PTWAddressL0	64	0	r-	current bank PTW address, level 0
PTWAddressL1	64	0	r-	current bank PTW address, level 1
PTWAddressL2	64	0	r-	current bank PTW address, level 2
PTWAddressL3	64	0	r-	current bank PTW address, level 3
PTWValueL0	64	0	r-	current bank PTW value, level 0
PTWValueL1	64	0	r-	current bank PTW value, level 1
PTWValueL2	64	0	r-	current bank PTW value, level 2
PTWValueL3	64	0	r-	current bank PTW value, level 3
PTWBase	64	0	r-	current bank PTW table base address
PTWInput	64	0	r-	current bank PTW input address
PTWOutput	64	0	r-	current bank PTW output address
PTWPgSize	64	0	r-	current bank PTW page size (Valid only when PTWOutput is valid)
PTWIE_EL1S	64	0	-w	perform secure EL1 stage 1 page table walk for fetch, filling PTW query registers
PTWD_EL1S	64	0	-w	perform secure EL1 stage 1 page table walk for load/store, filling PTW query registers
PTWIE_EL1NS	64	0	-w	perform non-secure EL1 stage 1 page table walk for fetch, filling PTW query registers
PTWD_EL1NS	64	0	-w	perform non-secure EL1 stage 1 page table walk for load/store, filling PTW query registers
PTWIE_EL2	64	0	-w	perform non-secure EL2 page table walk for fetch, filling PTW query registers
PTWD_EL2	64	0	-w	perform non-secure EL2 page table walk for load/store, filling PTW query registers
PTWIS_2	64	0	-w	perform non-secure stage 2 page table walk for fetch, filling PTW query registers
PTWD_S2	64	0	-w	perform non-secure stage 2 page table walk for load/store, filling PTW query registers
PTWIE_EL3	64	0	-w	perform EL3 page table walk for fetch, filling PTW query registers
PTWD_EL3	64	0	-w	perform EL3 page table walk for load/store, filling PTW query registers
PTWI_current	64	0	-w	perform current mode page table walk for fetch, filling PTW query registers
PTWD_current	64	0	-w	perform current mode page table walk for load/store, filling PTW query registers
ResetTLBs	8	0	-w	reset all implemented TLBs to initial state
HaltReason	8	0	r-	bit field indicating halt reason
atomicType	8	0	r-	current atomic instruction type (1:atomic, 2:exclusive)
WakeWFINow	8	0	-w	wake up core from WFI state
ambaResponse	64	0	rw	external AMBA-PV response

Table 13.25: Registers at level 2, type:CPU group:Integration_support

13.2.26 MPCore_distributor

Registers at level:2, type:CPU group:MPCore_distributor

Name	Bits	Initial-Hex	RW	Description
GICD_CIDR0	32	d	r-	Component ID 0
GICD_CIDR1	32	f0	r-	Component ID 1

GICD_CIDR2	32	5	r-	Component ID 2
GICD_CIDR3	32	b1	r-	Component ID 3
GICD_CLRSPLNSR	32	0	-w	Clear SPI
GICD_CLRSPLSR	32	0	-w	Clear SPI, Secure
GICD_CTLR	32	30	rw	Distributor Control
GICD_ICACTIVER0	32	0	rw	Interrupt Clear-Active 0
GICD_ICACTIVER1	32	0	rw	Interrupt Clear-Active 1
GICD_ICACTIVER2	32	0	rw	Interrupt Clear-Active 2
GICD_ICENABLER0	32	ffff	rw	Interrupt Clear-Enable 0
GICD_ICENABLER1	32	0	rw	Interrupt Clear-Enable 1
GICD_ICENABLER2	32	0	rw	Interrupt Clear-Enable 2
GICD_ICFGR0	32	aaaaaaaa	rw	Interrupt Configuration 0
GICD_ICFGR1	32	0	rw	Interrupt Configuration 1
GICD_ICFGR2	32	0	rw	Interrupt Configuration 2
GICD_ICFGR3	32	0	rw	Interrupt Configuration 3
GICD_ICFGR4	32	0	rw	Interrupt Configuration 4
GICD_ICFGR5	32	0	rw	Interrupt Configuration 5
GICD_ICPENDR0	32	0	rw	Interrupt Clear-Pending 0
GICD_ICPENDR1	32	0	rw	Interrupt Clear-Pending 1
GICD_ICPENDR2	32	0	rw	Interrupt Clear-Pending 2
GICD_IGROUPR0	32	0	rw	Interrupt Group 0
GICD_IGROUPR1	32	0	rw	Interrupt Group 1
GICD_IGROUPR2	32	0	rw	Interrupt Group 2
GICD_IGRPMODR0	32	0	rw	Interrupt Group Modifier 0
GICD_IGRPMODR1	32	0	rw	Interrupt Group Modifier 1
GICD_IGRPMODR2	32	0	rw	Interrupt Group Modifier 2
GICD_IIDR	32	102043b	r-	Distributor Implementor ID
GICD_IPRIORITYR0	32	0	rw	Interrupt Priority 0
GICD_IPRIORITYR1	32	0	rw	Interrupt Priority 1
GICD_IPRIORITYR2	32	0	rw	Interrupt Priority 2
GICD_IPRIORITYR3	32	0	rw	Interrupt Priority 3
GICD_IPRIORITYR4	32	0	rw	Interrupt Priority 4
GICD_IPRIORITYR5	32	0	rw	Interrupt Priority 5
GICD_IPRIORITYR6	32	0	rw	Interrupt Priority 6
GICD_IPRIORITYR7	32	0	rw	Interrupt Priority 7
GICD_IPRIORITYR8	32	0	rw	Interrupt Priority 8
GICD_IPRIORITYR9	32	0	rw	Interrupt Priority 9
GICD_IPRIORITYR10	32	0	rw	Interrupt Priority 10
GICD_IPRIORITYR11	32	0	rw	Interrupt Priority 11
GICD_IPRIORITYR12	32	0	rw	Interrupt Priority 12
GICD_IPRIORITYR13	32	0	rw	Interrupt Priority 13
GICD_IPRIORITYR14	32	0	rw	Interrupt Priority 14
GICD_IPRIORITYR15	32	0	rw	Interrupt Priority 15
GICD_IPRIORITYR16	32	0	rw	Interrupt Priority 16
GICD_IPRIORITYR17	32	0	rw	Interrupt Priority 17
GICD_IPRIORITYR18	32	0	rw	Interrupt Priority 18
GICD_IPRIORITYR19	32	0	rw	Interrupt Priority 19
GICD_IPRIORITYR20	32	0	rw	Interrupt Priority 20
GICD_IPRIORITYR21	32	0	rw	Interrupt Priority 21
GICD_IPRIORITYR22	32	0	rw	Interrupt Priority 22
GICD_IPRIORITYR23	32	0	rw	Interrupt Priority 23
GICD_IROUTER32	64	0	rw	Interrupt Routing 32
GICD_IROUTER33	64	0	rw	Interrupt Routing 33
GICD_IROUTER34	64	0	rw	Interrupt Routing 34
GICD_IROUTER35	64	0	rw	Interrupt Routing 35
GICD_IROUTER36	64	0	rw	Interrupt Routing 36

GICD_IROUTER37	64	0	rw	Interrupt Routing 37
GICD_IROUTER38	64	0	rw	Interrupt Routing 38
GICD_IROUTER39	64	0	rw	Interrupt Routing 39
GICD_IROUTER40	64	0	rw	Interrupt Routing 40
GICD_IROUTER41	64	0	rw	Interrupt Routing 41
GICD_IROUTER42	64	0	rw	Interrupt Routing 42
GICD_IROUTER43	64	0	rw	Interrupt Routing 43
GICD_IROUTER44	64	0	rw	Interrupt Routing 44
GICD_IROUTER45	64	0	rw	Interrupt Routing 45
GICD_IROUTER46	64	0	rw	Interrupt Routing 46
GICD_IROUTER47	64	0	rw	Interrupt Routing 47
GICD_IROUTER48	64	0	rw	Interrupt Routing 48
GICD_IROUTER49	64	0	rw	Interrupt Routing 49
GICD_IROUTER50	64	0	rw	Interrupt Routing 50
GICD_IROUTER51	64	0	rw	Interrupt Routing 51
GICD_IROUTER52	64	0	rw	Interrupt Routing 52
GICD_IROUTER53	64	0	rw	Interrupt Routing 53
GICD_IROUTER54	64	0	rw	Interrupt Routing 54
GICD_IROUTER55	64	0	rw	Interrupt Routing 55
GICD_IROUTER56	64	0	rw	Interrupt Routing 56
GICD_IROUTER57	64	0	rw	Interrupt Routing 57
GICD_IROUTER58	64	0	rw	Interrupt Routing 58
GICD_IROUTER59	64	0	rw	Interrupt Routing 59
GICD_IROUTER60	64	0	rw	Interrupt Routing 60
GICD_IROUTER61	64	0	rw	Interrupt Routing 61
GICD_IROUTER62	64	0	rw	Interrupt Routing 62
GICD_IROUTER63	64	0	rw	Interrupt Routing 63
GICD_IROUTER64	64	0	rw	Interrupt Routing 64
GICD_IROUTER65	64	0	rw	Interrupt Routing 65
GICD_IROUTER66	64	0	rw	Interrupt Routing 66
GICD_IROUTER67	64	0	rw	Interrupt Routing 67
GICD_IROUTER68	64	0	rw	Interrupt Routing 68
GICD_IROUTER69	64	0	rw	Interrupt Routing 69
GICD_IROUTER70	64	0	rw	Interrupt Routing 70
GICD_IROUTER71	64	0	rw	Interrupt Routing 71
GICD_IROUTER72	64	0	rw	Interrupt Routing 72
GICD_IROUTER73	64	0	rw	Interrupt Routing 73
GICD_IROUTER74	64	0	rw	Interrupt Routing 74
GICD_IROUTER75	64	0	rw	Interrupt Routing 75
GICD_IROUTER76	64	0	rw	Interrupt Routing 76
GICD_IROUTER77	64	0	rw	Interrupt Routing 77
GICD_IROUTER78	64	0	rw	Interrupt Routing 78
GICD_IROUTER79	64	0	rw	Interrupt Routing 79
GICD_IROUTER80	64	0	rw	Interrupt Routing 80
GICD_IROUTER81	64	0	rw	Interrupt Routing 81
GICD_IROUTER82	64	0	rw	Interrupt Routing 82
GICD_IROUTER83	64	0	rw	Interrupt Routing 83
GICD_IROUTER84	64	0	rw	Interrupt Routing 84
GICD_IROUTER85	64	0	rw	Interrupt Routing 85
GICD_IROUTER86	64	0	rw	Interrupt Routing 86
GICD_IROUTER87	64	0	rw	Interrupt Routing 87
GICD_IROUTER88	64	0	rw	Interrupt Routing 88
GICD_IROUTER89	64	0	rw	Interrupt Routing 89
GICD_IROUTER90	64	0	rw	Interrupt Routing 90
GICD_IROUTER91	64	0	rw	Interrupt Routing 91
GICD_IROUTER92	64	0	rw	Interrupt Routing 92

GICD_IROUTER93	64	0	rw	Interrupt Routing 93
GICD_IROUTER94	64	0	rw	Interrupt Routing 94
GICD_IROUTER95	64	0	rw	Interrupt Routing 95
GICD_ISACTIVER0	32	0	rw	Interrupt Set-Active 0
GICD_ISACTIVER1	32	0	rw	Interrupt Set-Active 1
GICD_ISACTIVER2	32	0	rw	Interrupt Set-Active 2
GICD_ISENABLER0	32	fff	rw	Interrupt Set-Enable 0
GICD_ISENABLER1	32	0	rw	Interrupt Set-Enable 1
GICD_ISENABLER2	32	0	rw	Interrupt Set-Enable 2
GICD_ISPENDR0	32	0	rw	Interrupt Set-Pending 0
GICD_ISPENDR1	32	0	rw	Interrupt Set-Pending 1
GICD_ISPENDR2	32	0	rw	Interrupt Set-Pending 2
GICD_NSACR0	32	0	rw	Interrupt Non-secure Access Control 0
GICD_NSACR1	32	0	rw	Interrupt Non-secure Access Control 1
GICD_NSACR2	32	0	rw	Interrupt Non-secure Access Control 2
GICD_NSACR3	32	0	rw	Interrupt Non-secure Access Control 3
GICD_NSACR4	32	0	rw	Interrupt Non-secure Access Control 4
GICD_NSACR5	32	0	rw	Interrupt Non-secure Access Control 5
GICD_PIDR0	32	92	r-	Peripheral ID 0
GICD_PIDR1	32	b4	r-	Peripheral ID 1
GICD_PIDR2	32	3b	r-	Peripheral ID 2
GICD_PIDR3	32	0	r-	Peripheral ID 3
GICD_PIDR4	32	44	r-	Peripheral ID 4
GICD_PIDR5	32	0	r-	Peripheral ID 5
GICD_PIDR6	32	0	r-	Peripheral ID 6
GICD_PIDR7	32	0	r-	Peripheral ID 7
GICD_SETSPI_NSR	32	0	-w	Set SPI
GICD_SETSPI_SR	32	0	-w	Set SPI, Secure
GICD_SPISR0	32	0	r-	SPI Status 0
GICD_SPISR1	32	0	r-	SPI Status 1
GICD_STATUSR	32	0	rw	Distributor Status
GICD_TYPER	32	7b0402	r-	Interrupt Controller Type

Table 13.26: Registers at level 2, type:CPU group:MPCore_distributor

13.2.27 MPCore_physical_redistributor

Registers at level:2, type:CPU group:MPCore_physical_redistributor

Name	Bits	Initial-Hex	RW	Description
GICR_CIDR0	32	d	r-	Redistributor Component ID 0
GICR_CIDR1	32	f0	r-	Redistributor Component ID 1
GICR_CIDR2	32	5	r-	Redistributor Component ID 2
GICR_CIDR3	32	b1	r-	Redistributor Component ID 3
GICR_CLRLPIR	64	0	-w	Clear LPI Pending
GICR_CTLR	32	0	rw	Redistributor Control
GICR_ICACTIVER0	32	0	rw	Interrupt Clear-Active 0
GICR_ICENABLER0	32	fff	rw	Interrupt Clear-Enable 0
GICR_ICFGR0	32	aaaaaaaa	rw	Interrupt Configuration 0
GICR_ICFGR1	32	0	rw	Interrupt Configuration 1
GICR_ICPENDR0	32	0	rw	Interrupt Clear-Pending 0
GICR_IGROUPR0	32	0	rw	Interrupt Group 0
GICR_IGRPMODR0	32	0	rw	Interrupt Group Modifier 0
GICR_IIDR	32	43b	r-	Redistributor Implementer Identification
GICR_INVALLR	64	0	-w	Redistributor Invalidate All

GICR_INVLPIR	64	0	-w	Redistributor Invalidate LPI
GICR_IPRIORITYR0	32	0	rw	Interrupt Priority 0
GICR_IPRIORITYR1	32	0	rw	Interrupt Priority 1
GICR_IPRIORITYR2	32	0	rw	Interrupt Priority 2
GICR_IPRIORITYR3	32	0	rw	Interrupt Priority 3
GICR_IPRIORITYR4	32	0	rw	Interrupt Priority 4
GICR_IPRIORITYR5	32	0	rw	Interrupt Priority 5
GICR_IPRIORITYR6	32	0	rw	Interrupt Priority 6
GICR_IPRIORITYR7	32	0	rw	Interrupt Priority 7
GICR_ISACTIVER0	32	0	rw	Interrupt Set-Active 0
GICR_ISENABLER0	32	fff	rw	Interrupt Set-Enable 0
GICR_ISPENDR0	32	0	rw	Interrupt Set-Pending 0
GICR_NSACR	32	0	rw	Interrupt Non-secure Access Control
GICR_PENDBASER	64	0	rw	Redistributor LPI Pending Table Base Address
GICR_PIDR0	32	93	r-	Redistributor Peripheral ID 0
GICR_PIDR1	32	b4	r-	Redistributor Peripheral ID 1
GICR_PIDR2	32	3b	r-	Redistributor Peripheral ID 2
GICR_PIDR3	32	0	r-	Redistributor Peripheral ID 3
GICR_PIDR4	32	44	r-	Redistributor Peripheral ID 4
GICR_PIDR5	32	0	r-	Redistributor Peripheral ID 5
GICR_PIDR6	32	0	r-	Redistributor Peripheral ID 6
GICR_PIDR7	32	0	r-	Redistributor Peripheral ID 7
GICR_PROPBASER	64	0	rw	Redistributor Properties Base Address
GICR_SETLPIR	64	0	-w	Set LPI Pending
GICR_STATUSR	32	0	rw	Redistributor Status
GICR_SYNCR	32	0	r-	Redistributor Synchronize
GICR_TYPER	64	9	r-	Redistributor Type
GICR_WAKER	32	6	rw	Redistributor Wake

Table 13.27: Registers at level 2, type:CPU group:MPCore_physical_redistributor

13.2.28 MPCore_processor_interface

Registers at level:2, type:CPU group:MPCore_processor_interface

Name	Bits	Initial-Hex	RW	Description
GICC_ABPR	32	3	rw	Aliased Binary Point
GICC_AEOIR	32	0	-w	Aliased End of Interrupt
GICC_AHPPIR	32	3ff	r-	Aliased Highest Priority Pending Interrupt
GICC_AIAR	32	3ff	r-	Aliased Interrupt Acknowledge
GICC_APR0	32	0	rw	Active Priorities 0
GICC_BPR	32	2	rw	Binary Point
GICC_CTLR	32	0	rw	CPU Interface Control
GICC_DIR	32	0	-w	Deactivate Interrupt
GICC_EOIR	32	0	-w	End of Interrupt
GICC_HPPIR	32	3ff	r-	Highest Priority Pending Interrupt
GICC_IAR	32	3ff	r-	Interrupt Acknowledge
GICC_IIDR	32	4043b	r-	CPU Interface ID
GICC_NSAPR0	32	0	rw	Non-secure Active Priorities 0
GICC_PMR	32	0	rw	Interrupt Priority Mask
GICC_RPR	32	ff	r-	Running Priority
GICC_STATUSR	32	0	rw	CPU Interface Status

Table 13.28: Registers at level 2, type:CPU group:MPCore_processor_interface

13.2.29 MPCore_virtual_interface_control

Registers at level:2, type:CPU group:MPCore_virtual_interface_control

Name	Bits	Initial-Hex	RW	Description
GICH_APR0	32	0	rw	Active Priorities 0
GICH_EISR0	32	0	r-	End of Interrupt Status 0
GICH_ELRSR0	32	f	r-	Empty List Register Status 0
GICH_HCR	32	0	rw	Hypervisor Control
GICH_LR0	32	0	rw	List 0
GICH_LR1	32	0	rw	List 1
GICH_LR2	32	0	rw	List 2
GICH_LR3	32	0	rw	List 3
GICH_MISR	32	0	r-	Maintenance Interrupt Status
GICH_VMCR	32	4c0000	rw	Virtual Machine Control
GICH_VTR	32	90000003	r-	VGIC Type

Table 13.29: Registers at level 2, type:CPU group:MPCore_virtual_interface_control

13.2.30 MPCore_virtual_processor_interface

Registers at level:2, type:CPU group:MPCore_virtual_processor_interface

Name	Bits	Initial-Hex	RW	Description
GICV_ABPR	32	3	rw	VM Aliased Binary Point
GICV_AEOIR	32	0	-w	VM Aliased End of Interrupt
GICV_AHPPIR	32	3ff	r-	VM Aliased Highest Priority Pending Interrupt
GICV_AIAR	32	3ff	r-	VM Aliased Interrupt Acknowledge
GICV_APR0	32	0	rw	VM Active Priorities 0
GICV_BPR	32	2	rw	VM Binary Point
GICV_CTLR	32	0	rw	Virtual Machine Control
GICV_DIR	32	0	-w	VM Deactivate Interrupt
GICV_EOIR	32	0	-w	VM End of Interrupt
GICV_HPPIR	32	3ff	r-	VM Highest Priority Pending Interrupt
GICV_IAR	32	3ff	r-	VM Interrupt Acknowledge
GICV_IIDR	32	4043b	r-	VM CPU Interface ID
GICV_PMR	32	0	rw	VM Priority Mask
GICV_RPR	32	ff	r-	VM Running Priority
GICV_STATUSR	32	0	rw	VM CPU Interface Status
ICV_APR0_EL1	32	0	rw	VM Group 0 Active Priorities 0
ICV_APIR0_EL1	32	0	rw	VM Group 1 Active Priorities 0
ICV_BPR0_EL1	32	2	rw	VM Group 0 Binary Point
ICV_BPR1_EL1	32	3	rw	VM Group 1 Binary Point
ICV_CTLR_EL1	32	400	rw	Virtual Machine Control
ICV_DIR_EL1	32	0	-w	VM Deactivate Interrupt
ICV_EOIR0_EL1	32	0	-w	VM Group 0 End of Interrupt
ICV_EOIR1_EL1	32	0	-w	VM Group 1 End of Interrupt
ICV_HPPIR0_EL1	32	3ff	r-	VM Group 0 Highest Priority Pending Interrupt
ICV_HPPIR1_EL1	32	3ff	r-	VM Group 1 Highest Priority Pending Interrupt
ICV_IAR0_EL1	32	3ff	r-	VM Group 0 Interrupt Acknowledge
ICV_IAR1_EL1	32	3ff	r-	VM Group 1 Interrupt Acknowledge
ICV_IGRPEN0_EL1	32	0	rw	VM Group 0 Interrupt Enable
ICV_IGRPEN1_EL1	32	0	rw	VM Group 1 Interrupt Enable
ICV_PMR_EL1	32	0	rw	VM Priority Mask
ICV_RPR_EL1	32	ff	r-	VM Running Priority

Table 13.30: Registers at level 2, type:CPU group:MPCore_virtual_processor_interface

13.2.31 MPCore ITS

Registers at level:2, type:CPU group:MPCore ITS

Name	Bits	Initial-Hex	RW	Description
GITS_BASER0	64	0	rw	ITS Translation Table Descriptor 0
GITS_BASER1	64	0	rw	ITS Translation Table Descriptor 1
GITS_BASER2	64	0	rw	ITS Translation Table Descriptor 2
GITS_BASER3	64	0	rw	ITS Translation Table Descriptor 3
GITS_BASER4	64	0	rw	ITS Translation Table Descriptor 4
GITS_BASER5	64	0	rw	ITS Translation Table Descriptor 5
GITS_BASER6	64	0	rw	ITS Translation Table Descriptor 6
GITS_BASER7	64	0	rw	ITS Translation Table Descriptor 7
GITS_CBASER	64	0	rw	ITS Command Queue Descriptor
GITS_CREADR	64	0	rw	ITS Read
GITS_CTLR	32	80000000	rw	ITS Control
GITS_CWRITER	64	0	rw	ITS Write
GITS_IIDR	32	43b	r-	ITS Identification
GITS_TYPER	64	9ef79	r-	ITS Type

Table 13.31: Registers at level 2, type:CPU group:MPCore ITS